Tectonophysics 757 (2019) 36–57

Contents lists available at ScienceDirect

Tectonophysics

journal homepage: www.elsevier.com/locate/tecto

Tectonic evolution and regional setting of the Cuu Long Basin, Vietnam

William J. Schmidt[a,*], Bui Huy Hoang[a], James W. Handschy[a,b], Vu Trong Hai[a], Trinh Xuan Cuong[a], Nguyen Thanh Tung[a]

[a] Vietnam Petroleum Institute, Vietnam
[b] Indiana University, United States of America

ARTICLE INFO

Keywords:
Cuu Long Basin
Mae Ping Fault
Kampot Fold Belt

ABSTRACT

The Cuu Long Basin is situated off the southeastern coast of Vietnam. This study utilizes all available seismic data in the Cuu Long Basin, Vinh Chau Basin and coastal Mekong Delta to constrain the tectonic history of the area. The linkage between the Mae Ping Fault Zone and deformation in the Cuu Long Basin is also examined in detail. Two major deformational events are observed: a transtensional rifting event from the mid Eocene to mid-Early Oligocene (40–31 Ma) followed by transpression from mid-Early Oligocene to mid-Late Oligocene (31–25 Ma). The data strongly suggest that all the very straight major tributaries of the Mekong River Delta follow northwest trending faults that link into to the southwestern end of the Cuu Long Basin fault system and that the basin opened between where the Mekong Delta faults enter and a point of zero extension at the NE termination of the basin. The Mekong Delta Fault Zone exhibited left-lateral displacement during extension in the Cuu Long Basin and right-lateral displacement during younger transpression. The orientation, age and sense of displacement of the Mekong Delta Fault Zone are consistent with slip on the Mae Ping Fault Zone in Thailand and Cambodia, and hence are considered to be part of the same regional fault system. Based on a Chainat Duplex analogue and other data, the Mekong Delta and Mae Ping fault zones appear to be linked by a NNW-striking step-over zone, referred to here as the Mekong-Ton Le Sap step-over zone. This zone follows the eastern boundary of the Paleocene to Early Eocene Kampot Fold Belt. Based on structural and stratigraphic similarities, it appears the adjacent Vinh Chau Basin shares a common tectonic history with the Cuu Long Basin, but the linkage to the Mekong Delta Fault Zone is less clear.

1. Introduction

The Cuu Long Basin is a NE trending basin 75 km off the southeastern coast of Vietnam, outboard of the Mekong Delta. It is separated from the Nam Con Son Basin at the southern end of the South China Sea rift system by a NE-SW trending block of relatively undeformed crust known as the Con Son Swell (Fig. 1). In physiographical terms the Cuu Long Basin is part of the South China Sea, but the geologic history is distinctly different. This paper examines the details of the structural and depositional history of the Cuu Long Basin and surrounding areas to constrain the tectonic history in more detail than previously possible.

Tapponnier et al. (1986) originally proposed that the formation of the Cuu Long Basin was linked to the southeastward extrusion of Indochina caused by the India-Eurasia collision. Their original proposal (Tapponnier et al., 1986) has since been elaborated on by numerous authors. Most authors extend the Mae Ping Fault (also referred to as the Wang Chao Fault) from Thailand to the Cuu Long Basin (Tapponnier et al., 1986; Huchon et al., 1994; Lacassin et al., 1997; Leloup et al., 2001; Morley, 2002; Fyhn et al., 2009). Some terminate the Mae Ping Fault in the Cuu Long Basin (Tapponnier et al., 1986; Lacassin et al., 1997; Morley, 2002) while others extend the Mae Ping Fault southeastward to the NW Borneo margin (Peltzer and Tapponnier, 1988; Leloup et al., 2001; Replumaz and Tapponnier, 2003). Morley (2002) considered extension of the Mae Ping Fault to the NW Borneo margin to be unlikely because the Mae Ping Fault trend is not present in the Nam Con Son Basin. In addition, Morley (2002) points out that the Proto-South China Sea separated Indochina from Borneo in the Eocene-Oligocene and the Miocene-Pliocene fold-thrust belt offshore NW Borneo formed at the wrong time to be associated with the Mae Ping Fault. Lee et al. (2001) interpreted the curved fault pattern in the Cuu Long Basin to be the result of a rift zone that was controlled by pre-existing zones of weakness in the basement and did not link the Cuu Long Basin to the Mae Ping Fault Zone. In contrast, Morley (2002) considered termination of the Mae Ping Fault in the Cuu Long Basin as the best explanation for the fault pattern interpreted by Lee et al. (2001).

The challenge facing all of those interpretations is that the

* Corresponding author at: Vietnam Petroleum Institute (VPI), 5th floor, VPI Tower, 167 Trung Kinh Str., Yen Hoa Ward, Cau Giay Dist., Hanoi, Vietnam.
E-mail address: schmidt.wj@vpi.pvn.vn (W.J. Schmidt).

https://doi.org/10.1016/j.tecto.2019.03.001
Received 27 October 2018; Received in revised form 19 February 2019; Accepted 2 March 2019
Available online 05 March 2019
0040-1951/ © 2019 Elsevier B.V. All rights reserved.

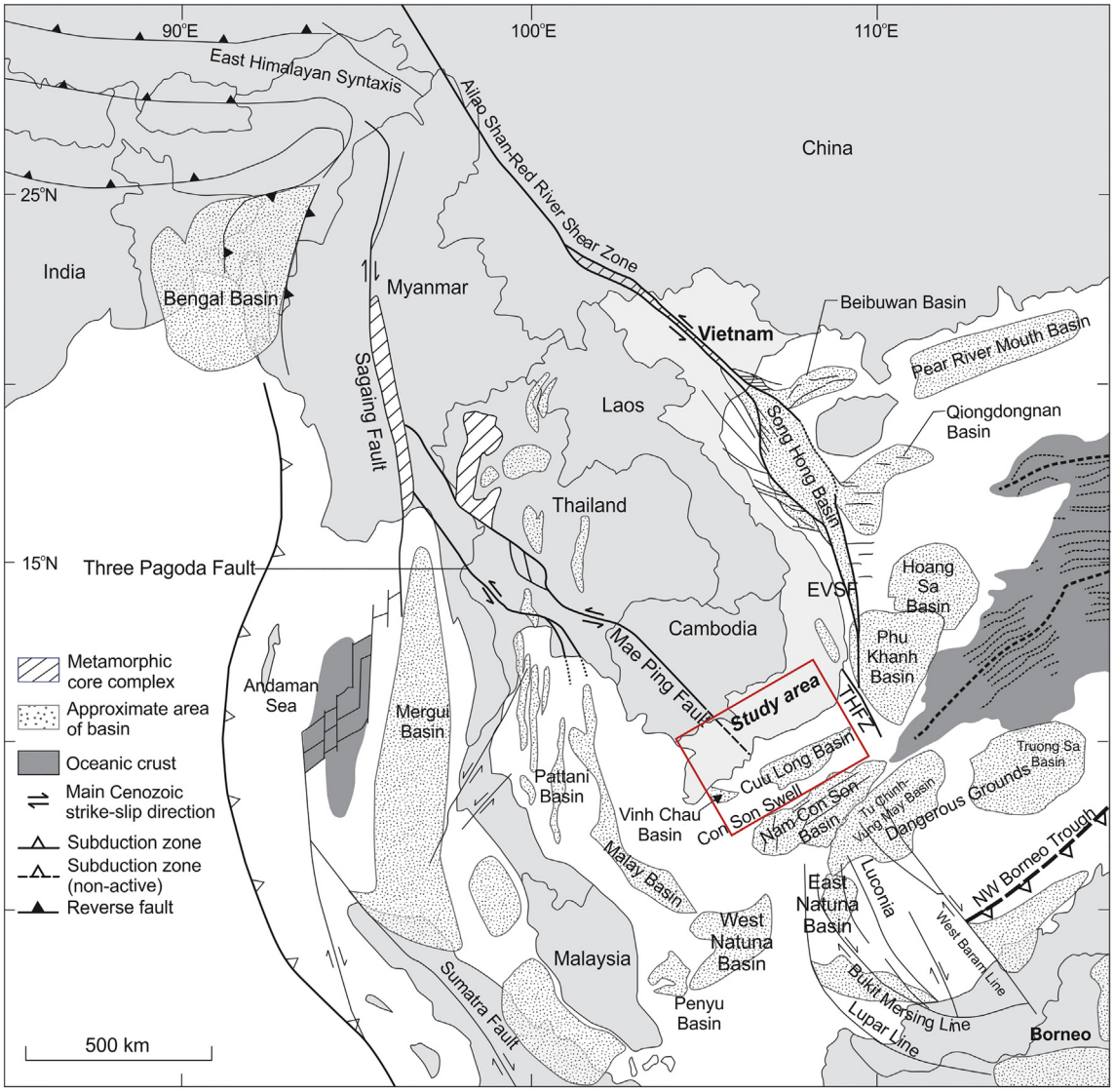


Fig. 1. Location of major structural features and Cenozoic basins for the greater Indochina area, after Fyhn et al. (2009). The study area, that includes the Cuu Long and Vinh Chau basins, in southern Vietnam is indicated.

continuation of the Mae Ping Fault into Vietnam must be inferred due to Quaternary sediments covering the area from Ton Le Sap Lake in Cambodia to the Vietnamese coast. Inferred Mae Ping Fault trends were projected to the Cuu Long Basin often along straight lines and in different locations. Prior to this study, published fault maps of the Cuu Long Basin were loosely constrained. Lee et al. (2001) made the most compelling case for fault trends in the basin based on a regional grid of 2D seismic lines with an average line spacing of 40–50 km. But even the Lee et al. (2001) dataset was too coarse to reliably correlate faults from line-to-line. More recently, Fyhn et al. (2009) used a seismic dataset with a tighter line spacing of 10–20 km, but it only covered the northernmost part of the basin.

To provide better constraints for the structure and deformational history in the Cuu Long Basin, this study used all the existing 2D and 3D seismic data in the basin and Mekong Delta plus stratigraphic data from 144 wells (Fig. 2). Where 3D seismic does not exist, the average 2D line spacing is < 10 km over most of the basin (Fig. 2). As a result, this paper provides better characterizations of the fault geometries, linkage between the Cuu Long Basin and the Mae Ping Fault, the ages of depositional sequences, and the deformation history in the basin.

2. Geologic setting

The Cuu Long Basin is a prolific oil and gas basin situated offshore from the southeastern coast of Vietnam (Fig. 1). It is located between the Vietnamese mainland and the Nam Con Son Basin at the southwestern end of the South China Sea. The Con Son Swell, a relatively unfaulted basement high, separates the Cuu Long Basin from the Nam Con Son Basin. Crustal extension in the Nam Con Son Basin is estimated to be about 5 times greater than in the Cuu Long Basin (Huchon et al., 1998). Extension in the Nam Con Son Basin is also younger than in the Cuu Long Basin and is closely related to rifting and seafloor spreading in the South China Sea (Matthews et al., 1997).

2.1. Tectonic evolution

Rifting, compression, and thermal sag subsidence have been recognized in the Cuu Long Basin in previous evaluations. Those authors noted that rifting started in the Eocene and continued to the Oligocene-Miocene boundary (Lee et al., 2001; Schmidt et al., 2003; Hung and Hung, 2004; Fyhn et al., 2009) or to the end of the Early Miocene (Cuong and Warren, 2009). Compression in the basin was not noted by Lee et al. (2001), but more recent publications have recognized and

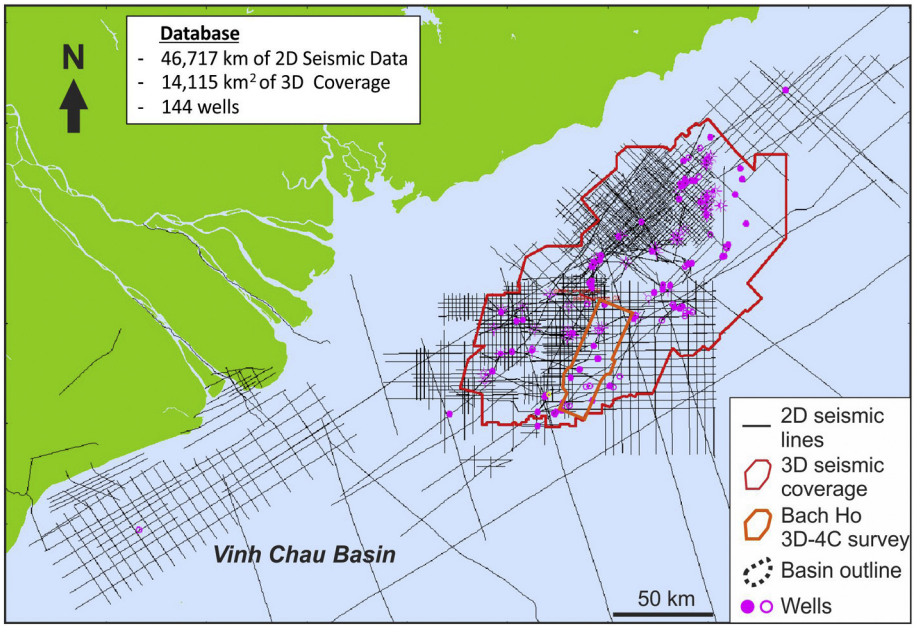


Fig. 2. Seismic and well data used for this paper includes the Cuu Long Basin, Vinh Chau Basin and coastal and onshore portions of the Mekong Delta. The 3D seismic surveys cover the majority of the Cuu Long Basin. The high quality 4C survey over Bach Ho and Rong fields is also shown.

dated it as Late Oligocene to Early Miocene (Schmidt et al., 2003; Hung and Hung, 2004; Cuong and Warren, 2009; Fyhn et al., 2009). The compression created folds and reverse faults, the most prominent of which occurs at the Bach Ho Field (Cuong and Warren, 2009). Onset of thermal sag subsidence was noted to begin in Early Miocene (Lee et al., 2001; Fyhn et al., 2009) or Middle Miocene (Cuong and Warren, 2009).

2.2. Stratigraphy

Previous publications such as Lee et al. (2001) and Fyhn et al. (2009) proposed seismic stratigraphic sequence definitions for the basin. Lee et al. (2001) derived their stratigraphy in the Cuu Long Basin from a single well and its correlation to the seismic sequences identified from regional 2D seismic lines with 40–50 km line spacing. Fyhn et al. (2009) only deals with the northern part of the Cuu Long Basin, and the stratigraphy is derived from three wells, which were correlated to a slightly better 2D seismic dataset. Both Lee and Fyhn use their own stratigraphy naming convention, which differs from the sequence subdivisions used by the oil and gas operators in the basin. This makes it difficult to take full advantage of all of the oil and gas data in the basin because most of the well-to-seismic ties come from the operators who drilled the wells. We have adopted the sequence stratigraphic scheme in general use by operators in the basin (Fig. 3). It consists of the pre-Tertiary crystalline basement plus six Cenozoic seismic-stratigraphic sequences, designated A–F. A brief description of the basement and overlying sequences follows.

- The basement is made up primarily of intrusive granitoids ranging in age from Middle Jurassic to Latest Cretaceous (Bao et al., 2001) with minor amounts of metamorphic host rocks. Similar granitoids are present onshore in Vietnam (Fyhn et al., 2010; Shellnutt et al., 2013).
- The Middle Eocene F sequence (sometimes referred to locally as the G sequence) is the oldest sedimentary sequence in the basin and is only present in the deepest parts of the basin. It consists of coarse-grained sandstone and conglomerate, with minor amounts of fine-grained silt and shale. It was deposited in high-energy alluvial and fluvial environments.
- The Late Eocene E sequence consists of coarse to fine grained sandstone plus lesser amounts of conglomerate that are interbedded with silt and shale. It was deposited in fluvial to swampy lacustrine environments.
- The Early Oligocene D sequence is dominated by a thick succession of fine-grained silt and shale, deposited in a deep-water lacustrine environment.
- The late Early Oligocene C sequence is sandier than the D sequence and consists of medium- to very fine-grained sandstone interbedded with silt and shale. Deposition of the C sequence was in lacustrine deltaic systems filling the final remnants of the rift basin lakes.
- The Late Oligocene to Miocene B sequence consists of siliciclastic sediments deposited in deltaic to open marine settings.
- The Pliocene-Quaternary A sequence consists of coarse-grained unconsolidated sand and shale with minor amounts of interbedded carbonate and coal layers deposited in marginal marine environments.

The ages for the siliciclastic sequences used in this paper (Fig. 3) are based on a 2018 high resolution biostratigraphic analysis carried out by R. Morley and Vietnam Petroleum Institute (VPI) staff (Morley et al., in preparation). The age of the Paleogene sequences from Morley et al. (in preparation) and Dam and Quang (2015) are about 4 Ma older than previous evaluations of the basin. Morley et al. (in preparation) used the same seismic horizons and well tops used in this paper. The dates from Morley et al. (under review) are better constrained than previous evaluations (e.g., Tran and Phung, 2007), but some uncertainty about the absolute ages of the sequences still remains. Deformational features in this study are described with respect to the depositional sequences they deform, irrespective of the absolute ages.

3. Data and methodology

3.1. Database

The database for this study includes 144 exploration wells, > 46,700 km of good quality 2D seismic reflection data covering the Cuu Long Basin, Vinh Chau Basin and coastal portions of the Mekong Delta, and > 14,000 km^2 of 3D seismic reflection data (Fig. 2). These seismic data are the result of > 40 years of exploration work in the region by

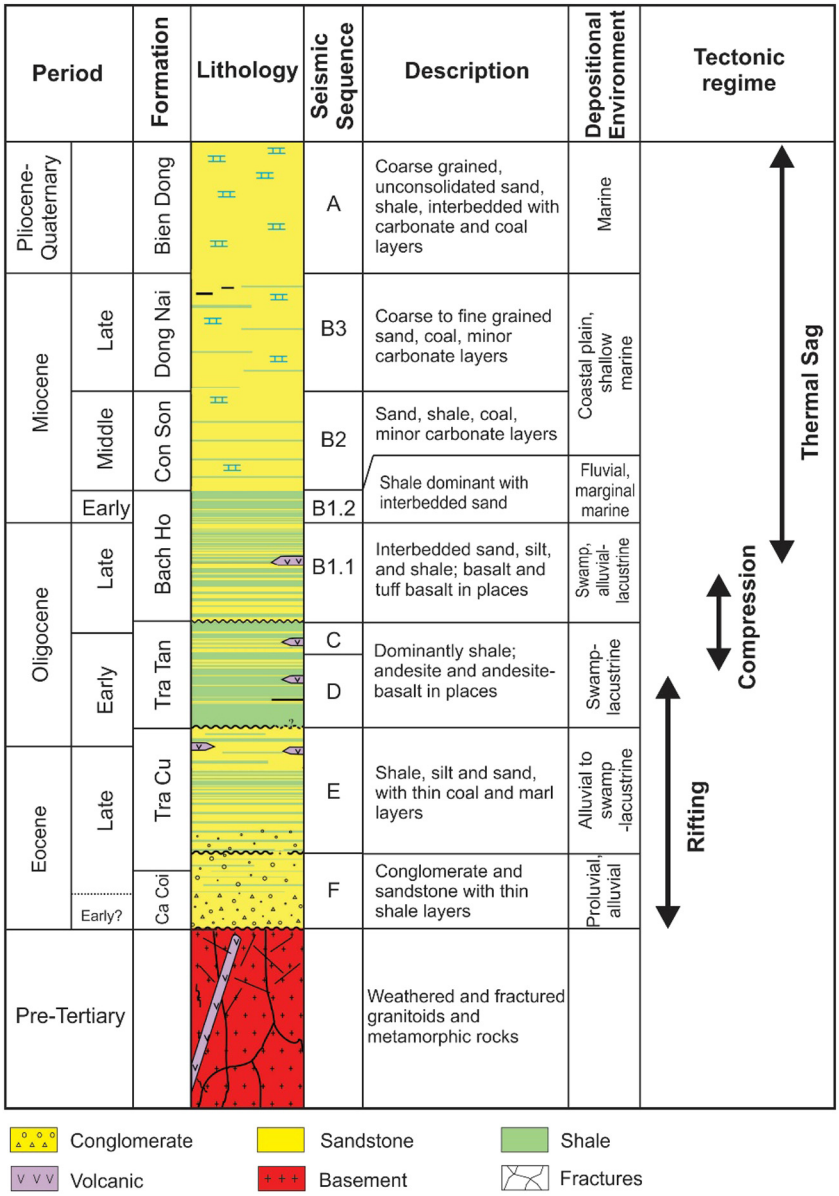


Fig. 3. Cuu Long Basin stratigraphic column with associated depositional environments and tectonic regimes. The biostratigraphic ages are from Morley et al. (under review).
Figure is modified from Tran and Phung (2007).

several operators. The seismic data have variable acquisition parameters and have been processed using a variety of techniques. Recently, most of the 3D seismic surveys were reprocessed using advanced techniques such as Pre-Stack Depth Migration and Controlled Beam Migration, resulting in much better data quality. At Bach Ho and Rong fields, a 4-component 3D seismic survey (Fig. 2), acquired in 2015 with multi-component seafloor receivers, yielded high quality imaging of the complex structures. It provides much greater detail about fault geometries and timing of deformational events than was possible with the older data. The seismic data have been tied to the well data shown in Fig. 2 using check-shot and/or seismic interval velocities. In aggregate, this dataset contains orders of magnitude more data than used for previously published Cuu Long Basin studies.

The integration and interpretation of the well and seismic datasets were conducted during several proprietary studies, either internal VPI evaluations or as joint studies by VPI and third parties. The latest of these was done in cooperation with R. Morley to integrate high resolution biostratigraphic analysis across the Cuu Long Basin (Morley et al., under review). Proprietary regional gravity data, recently published research in adjacent basins (e.g., Fyhn et al., 2010, 2016; Morley, 2012, 2013), published geologic maps (e.g., Tien et al., 2009), and satellite images of Vietnam and Cambodia were also used.

3.2. Structural and stratigraphic maps

The seismic dataset was used to interpret the top of the crystalline basement and five major Tertiary clastic sequence boundaries (Top F, Top E, Top D, Top C, and Top B1; Fig. 3). These sequence boundaries are shown in the stratigraphic column for the Cuu Long Basin (Fig. 3) and have been tied to all 144 wells in the database (Fig. 2). While there is a small amount of uncertainty about the absolute ages of some stratigraphic units due to the limitations of biostratigraphic age dating of non-marine sediments, the sequence boundaries correlate very well across the entire basin and there is very little mapping uncertainty for each sequence boundary. Five depth structure maps were made on: 1) the top of the acoustic basement; 2) the top of the combined E and F

sequences (denoted as E/F); 3) the top of the D sequence; 4) the top of the C sequence; and 5) the top of the B1 sequence. Isopach maps were constructed for the E/F, D, and C sequences using the depth maps.

Seismic mapping in the Vinh Chau Basin utilized a 3 km × 5 km 2D seismic grid that covered the entire basin, including the transition between NW faults and NE faults that occurs just offshore from the coastline (Fig. 2). Biostratigraphic ages from one well in the Vinh Chau Basin were used to constrain the ages of the mapped seismic horizons in the basin.

4. Data interpretation and analysis

The following sections provide an overview of the results of the seismic interpretation, mapping and other analyses.

4.1. Seismic interpretation

The seismic characteristics of the six interpreted stratigraphic horizons are as follows:

- The Top Basement seismic pick is characterized by a high amplitude, low frequency reflector, and has fair to good continuity on basin flanks and on basement highs. In deeper parts of the basin the Top Basement pick is less obvious due to less acoustic impedance contrast between the strongly lithified E and F sequence rocks and the basement. Deformation due to faulting and folding also makes the interpretation more difficult on the flanks of some basement highs.
- The Top F seismic pick is usually a strong amplitude reflector. Its continuity is variable from poor to good and it is often difficult to correlate from area to area. Internally, the F seismic sequence has low amplitude and chaotic reflections. The Top F is very difficult to interpret across the entire basin due to the lack of well control and because strong compaction in the deeper basins greatly diminishes the acoustic impedance contrast between the E and F sequence. As a result, the Top F is considered tentative.
- The Top E seismic reflector has a strong amplitude response, but poor continuity. It is a basin-scale regional unconformity that forms the boundary between the underlying E sequence, which has strong reflectivity with poor to good continuity, and the overlying D sequence which has low to moderate reflectivity and moderate to high reflector continuity. Truncation of internal E sequence reflections, onlap of the D sequence and overlapping sub-parallel reflections of the D sequence on the Top E horizon are commonly observed.
- The Top D is a weak and discontinuous reflector. A clear top-lap geometry is widely observed on the basin flanks. Internally, the D sequence has very little reflectivity across most the basin. High amplitude reflections in some localized areas correspond to the presence of volcanic interbeds.
- The Top C is characterized by a low frequency, low to high amplitude, and moderate continuity reflector. It onlaps the basin flanks and frequently onlaps the crests of anticlines. Internally, the C sequence has moderate to high amplitude reflectors.
- The Top B1 has a high amplitude, high frequency, and good continuity reflector. It can be correlated with high confidence across the entire basin.

4.2. Basement faulting in the Mekong Delta and offshore basins

The basement depth structure map for the offshore Cuu Long and Vinh Chau basins with a regional geologic map of the adjacent onshore area are shown in Fig. 4. The top of the pre-Tertiary basement reaches depths up to 10 km in the Cuu Long Basin and approximately 4 km in the Vinh Chau Basin (Fig. 4). Water depths average < 100 m in both basins. Near shore and onshore 2D seismic data (Fig. 2) made it possible to extrapolate faults from the offshore basins to the coast. Faults mapped along and immediately offshore from the coastal tributaries of the Mekong River strike NW-SE. Onshore seismic lines are generally parallel to the faults seen immediately offshore and do not show much faulting, presumably because they are parallel to the onshore continuations of the offshore faults. Offshore the faults gradually change strike and curve counter-clockwise into the NE-striking faults in the Cuu Long Basin. The same NW to NE counter-clockwise curvature of fault trends is observed within the Vinh Chau Basin (Fig. 4). There is no indication that any of the NW striking faults extend across the Con Son Swell (Fig. 1) to the southeast.

The upward terminations of faults mapped immediately offshore of the Mekong Delta coincide almost perfectly with the outlets of Mekong River tributaries at the coastline. Upstream these Mekong River tributaries follow very linear courses (Fig. 5A). This is markedly different from the meandering course of the Saigon River to the NE (Fig. 5B). Both the Mekong and Saigon deltas are located on flat, low-lying flood plains. The Mekong River tributaries imply linear structural controls for the river channels while the Saigon River channels do not. Therefore, we infer that the linear faults mapped near the mouths of the Mekong River tributaries extend to the NW along the linear river channels (Fig. 4). We refer to these mapped and inferred faults as the Mekong Delta Fault Zone. The main axis of the Cuu Long Basin trends NE, orthogonal to the NW strike of the Mekong Delta Fault Zone (Fig. 4).

4.3. Structural analysis and tectonic regimes

Two regional seismic lines in Fig. 6 illustrate the seismic reflection events that have been mapped throughout the Cuu Long Basin. Faults and folds active during deposition of the mapped sequences were documented and provide constraints on the ages of deformation in the basin. Summary plots of structural trends were analyzed by age and subdivided into structural domains within the Cuu Long Basin. The strikes of faults and fold axial traces were sampled every 2 km and then plotted on rose diagrams.

Three tectonic regimes are evident from the seismic interpretation: 1) rifting (normal faults), 2) compression (reverse faults and folds) and, 3) thermal sag (relatively unfaulted basin fill sequences). These tectonic regimes are illustrated on the regional seismic sections shown in Fig. 6. A regional seismic line from the Vinh Chau Basin (Fig. 7) shows similar stratigraphy and tectonic regimes. Structures associated with each tectonic regime are discussed below. The ages of the tectonic regimes are illustrated in Figs. 3, 6 and 7.

4.3.1. Rifting

The first phase of deformation in the Cuu Long Basin was rifting along NE- to NNE-striking normal faults. It created the NE-SW trend of the basin axis seen on the top basement depth map (Fig. 4). Rifting took place during deposition of the Eocene to Early Oligocene F, E, and D sequences (Fig. 6). Faulting ended during deposition of the uppermost D sequence (Fig. 6). Stratigraphic offsets on the large syn-depositional normal faults were used to date the age of the rifting event (Fig. 6). We estimate rifting stopped in the uppermost D sequence around 31 Ma. A few of the larger faults appear to have normal offset continuing into the lower part of the C sequence, but this is the result of differential compaction, not continued extension during C sequence deposition.

Regional dip lines (Fig. 6) and isopach maps (Fig. 8) show that the combined E/F sequence and the D sequence both thicken into the central part of the basin. The overall maximum isopach thickness trends NE-SW, similar to the trend of the basin axis on the top basement (Fig. 4). Within the E/F isopach, NNE-trending sub-basins are related to horsts and grabens that formed during deposition of these sequences. The contact between the E and F sequences is conformable in the grabens, but on the flanks of basement highs the F sequence onlaps the highs or is separated from the E sequence by an unconformity. The Early Oligocene D sequence onlaps a regional unconformity on the top of the E sequence across most of the basin (Fig. 6). Throw on the rift-related normal faults decreases, often to zero, in the upper portion of

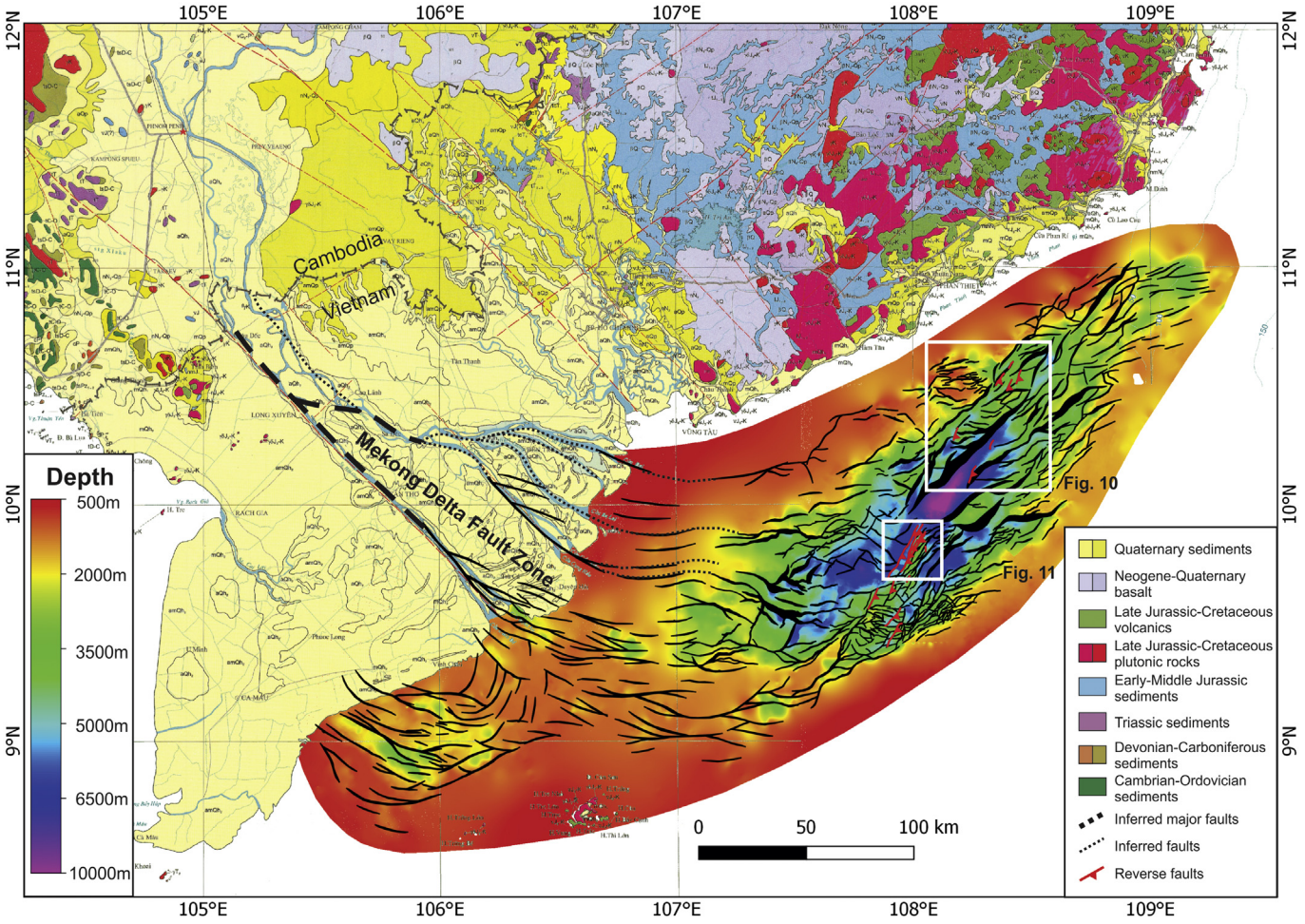


Fig. 4. Geology of onshore areas (Tien et al., 2009) adjacent to the Cuu Long and Vinh Chau basins. The offshore area shows the depth and structure at the top of basement. Mekong Delta faults trend NW, at 90° to the NE trend of the axis of the Cuu Long Basin. The inset location maps for Figs. 10 and 11 are shown.

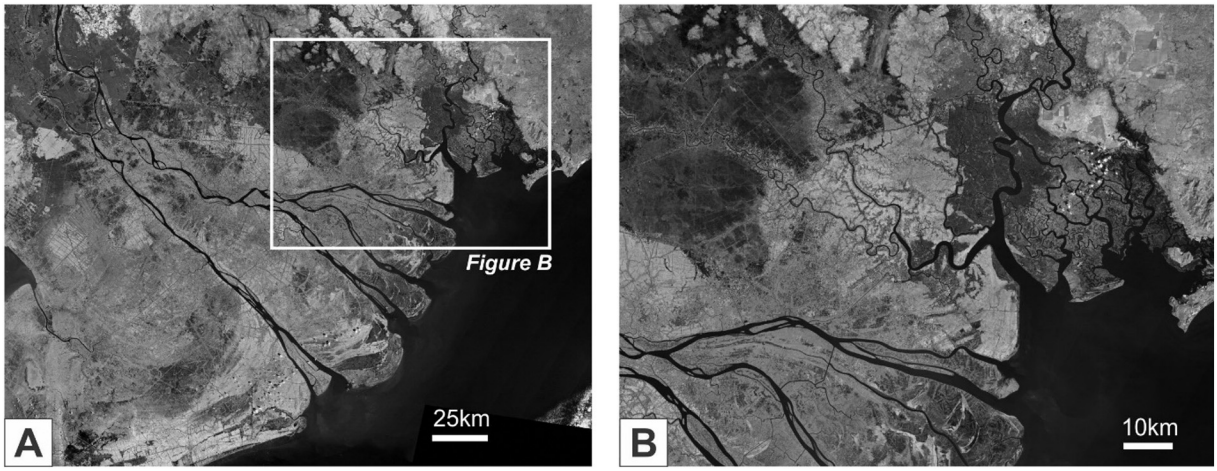


Fig. 5. Satellite images of the Mekong River Delta in South Vietnam (A) and a close up of the Saigon River area to the northwest (B). There is a marked difference between the meandering Saigon River complex versus the linear course of the Mekong River tributaries. It is inferred from these and other data, that the Mekong River tributaries coincide with subsurface faults, as shown in Fig. 4.

the D sequence, just prior to the onset of compressional folding in the uppermost D and C sequences. Because of the upward diminishing throw on the faults, the structural detail on the D isopach is less pronounced than on the E/F isopach. Fig. 9 shows maps of structures in the E/F, D, C and B1 sequences.

The 3D seismic sections in Figs. 10 and 11 provide examples of faulting and depositional contacts observed near basement highs. Syndepositional fault-block rotation, footwall uplift, and erosional truncation created local unconformities over horst blocks within the E and F sequences (Fig. 10C). Three seismic sections from the northern flank to the crest of Bach Ho Field are shown in Fig. 11. North of the field the E sequence overlies basement (Fig. 11A), it thins up structure to the south

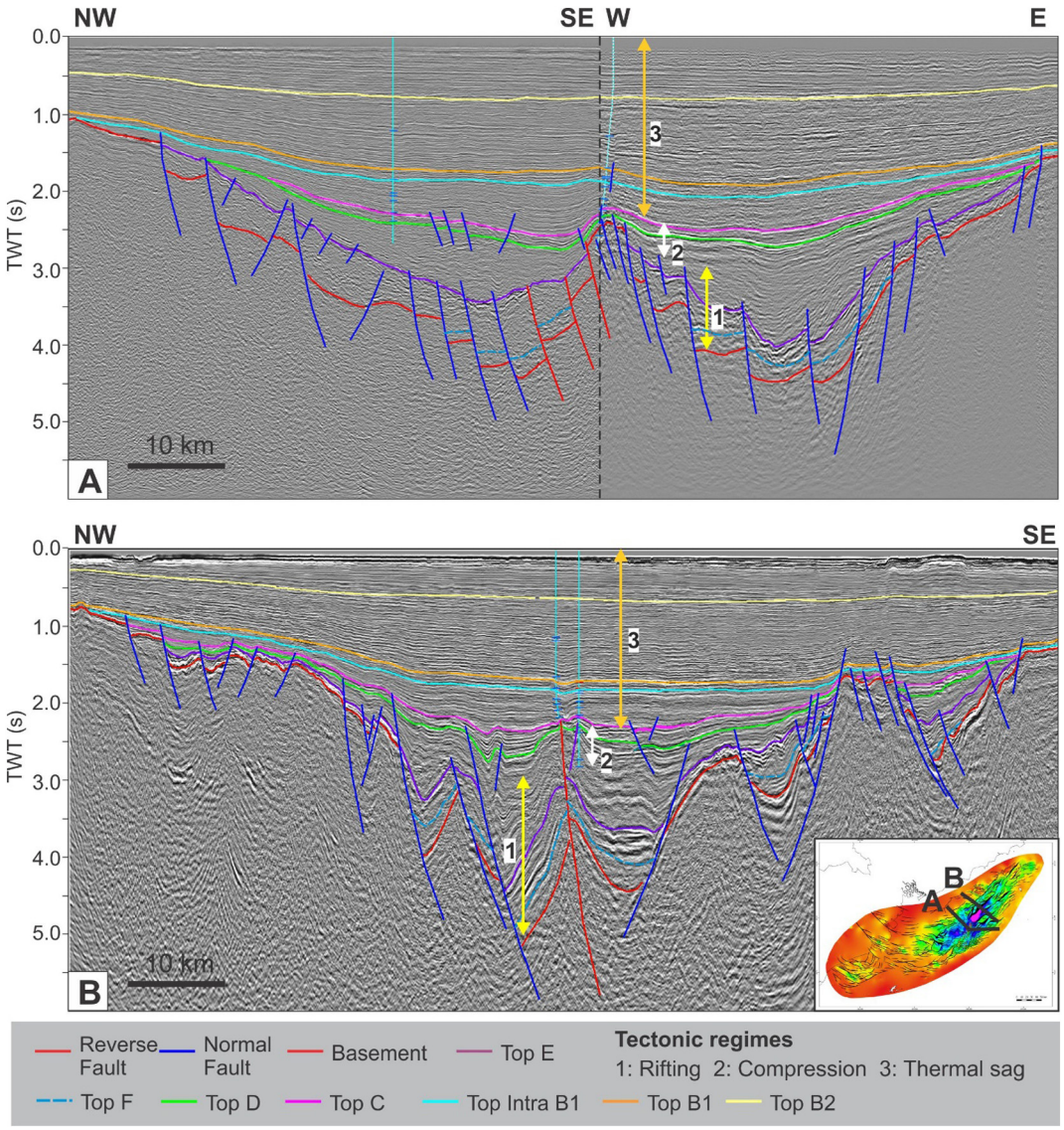


Fig. 6. Interpreted regional seismic lines across the central part of the Cuu Long Basin. The age of rifting, compression and thermal sag regimes are indicated. This relationship is seen throughout the basin and is used to define the end of rifting in the first tectonic regime. The vertical axis in the sections is displayed in two-way travel time.

(Fig. 11B) and is absent at the crest of the field (Fig. 11C). The younger compressional features on these seismic lines are discussed in the following section.

The trends of all mapped top basement faults for the Cuu Long Basin and Mekong Delta have been plotted in 4 domains (Fig. 12).

- Domain 1 is the area from the onshore Mekong Delta to the SW corner of the Cuu Long Basin;
- Domain 2 is the NW flank of the basin;
- Domain 3 is the central, deep axis of the basin;
- Domain 4 is the SE flank of the basin.

The main NW-striking Mekong Delta faults near the coast curve counter-clockwise in Domain 1 and the SW part of Domain 2 as they enter the Cuu Long Basin. This is consistent with the observations of Lee et al. (2001). Domain 3 contains the largest NE-striking faults, a large number of NNE-striking faults and a few ENE-striking faults (Fig. 12). The NNE-striking faults are along trend with and parallel to the NNE-striking Paleocene to mid-Eocene dikes mapped onshore (Fig. 12). The trend of faults in Domain 4 is quite similar to that of Domain 2 (Fig. 12); both of these domains flank the central axis of the basin (Domain 3).

4.3.2. *Compression*

The transition between rifting and compression occurred during deposition of the uppermost D sequence (Figs. 3 and 6). The fault pattern for the E/F sequence (Fig. 9) is primarily related to rifting; there is no evidence of compression in this time interval. Lower and mid-D sequence deposition was dominated by normal fault generated accommodation space. Fault throw diminishes in the upper D sequence just prior to or at the same time as evidence for NNE-trending compressional folds and reverse faults begin to appear (Fig. 9). Compressional features continued to form during deposition of the C and lowermost B1 sequences. Axial traces for compressional folds in the D, C and B1 sequences trend NNE, parallel to inverted NNE normal faults and basement involved reverse faults of the same age (Fig. 9), and parallel onshore Paleocene dike swarms (Fig. 12). The compressional folds and reverse faults are mainly formed in the central portion of the basin where the greatest amount of extension had previously occurred (Domain 3, Fig. 12).

Folding and reverse faulting in the upper D sequence is seen on

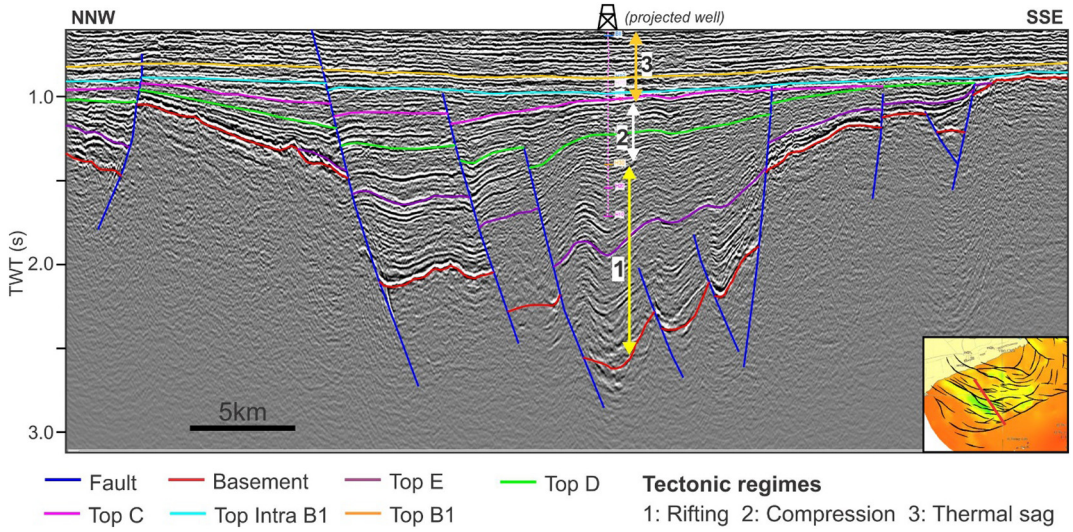


Fig. 7. 2D seismic section from the Vinh Chau Basin. Like in the Cuu Long Basin, the main rift phase takes place during deposition of the E/F sequence and most of the D sequence. Rifting is followed by a compressional event that mainly folds the E and D sequences. Some of the displacement on normal faults that offset the top of D and C is related to differential compaction. The top of B1 is nearly unfaulted.

seismic sections from Bach Ho Field (Fig. 11). At the northern end of Bach Ho Field there is clear evidence of reverse faulting and associated folding in the E, D, and C sequences (Fig. 11A). The magnitude of the basement reverse faulting increases to the south (Fig. 11B). The largest reverse throw in the basin is at the crest of Bach Ho Field, where the basement was thrust over E and D sequence rocks (Fig. 11C). This intense reverse faulting at Bach Ho Field corresponds to the highest basement fracture porosity in the field (Cuong and Warren, 2009). The structural relief and deformational intensity make Bach Ho Field the largest fractured basement field in the Cuu Long Basin, with cumulative production from the basement of > 1.2 billion barrels of oil (scout reports).

Basement reverse faulting at Su Tu Nau Field (Fig. 10A and B) also occurred during deposition of the uppermost D sequence. Reverse faults at the crest of Su Tu Nau Field offset older normal faults (Fig. 10A). These normal faults were active just prior to the late D inversion (Fig. 10A). The equivalent normal faults are not offset by reverse faults on the southern flank of Su Tu Nau, where compression is less severe (Fig. 10B).

Reverse faulting inversion of the older normal faults often has associated folding in the overlying siliciclastic sequences (Fig. 10A). At Su Tu Trang Field, inversion created a large compressional fold in the D sequence on the northwest flank of the field (Fig. 10C). The crest of the compressional anticline corresponds to the thickest part of the extensional D sequence depocenter and the fold was set up by reverse fault reactivation of the former normal fault. The overlying C sequence beds onlap the folded D sequence, indicating that folding at Su Tu Trang ended just before or early during deposition of the C sequence. Offset of the top D and C sequences on small normal faults over the crest of Su Tu Trang Field is most likely related to differential compaction after the folding of the D and lower C sequences.

Examples of intraformational D and C sequence folds are shown in Fig. 13. The thicknesses of younger sediments increase away from the hinge lines of these folds (Fig. 13C), indicating fold growth during deposition. In addition, fold limb dips flatten, and fold amplitude is lower in the overlying C sequence (Fig. 13C). Two intra-D horizons in

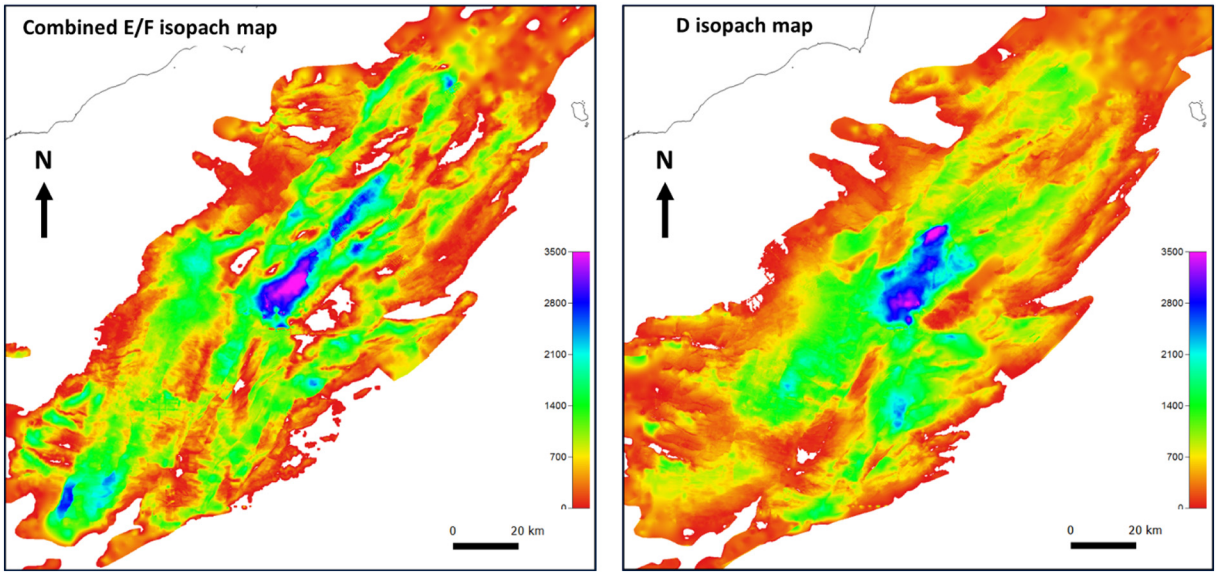


Fig. 8. Isopach maps for the combined E/F and D sequences. The NE trend of the isopach maps is similar to the trend of basin axis on the top of basement depth map (Fig. 4).

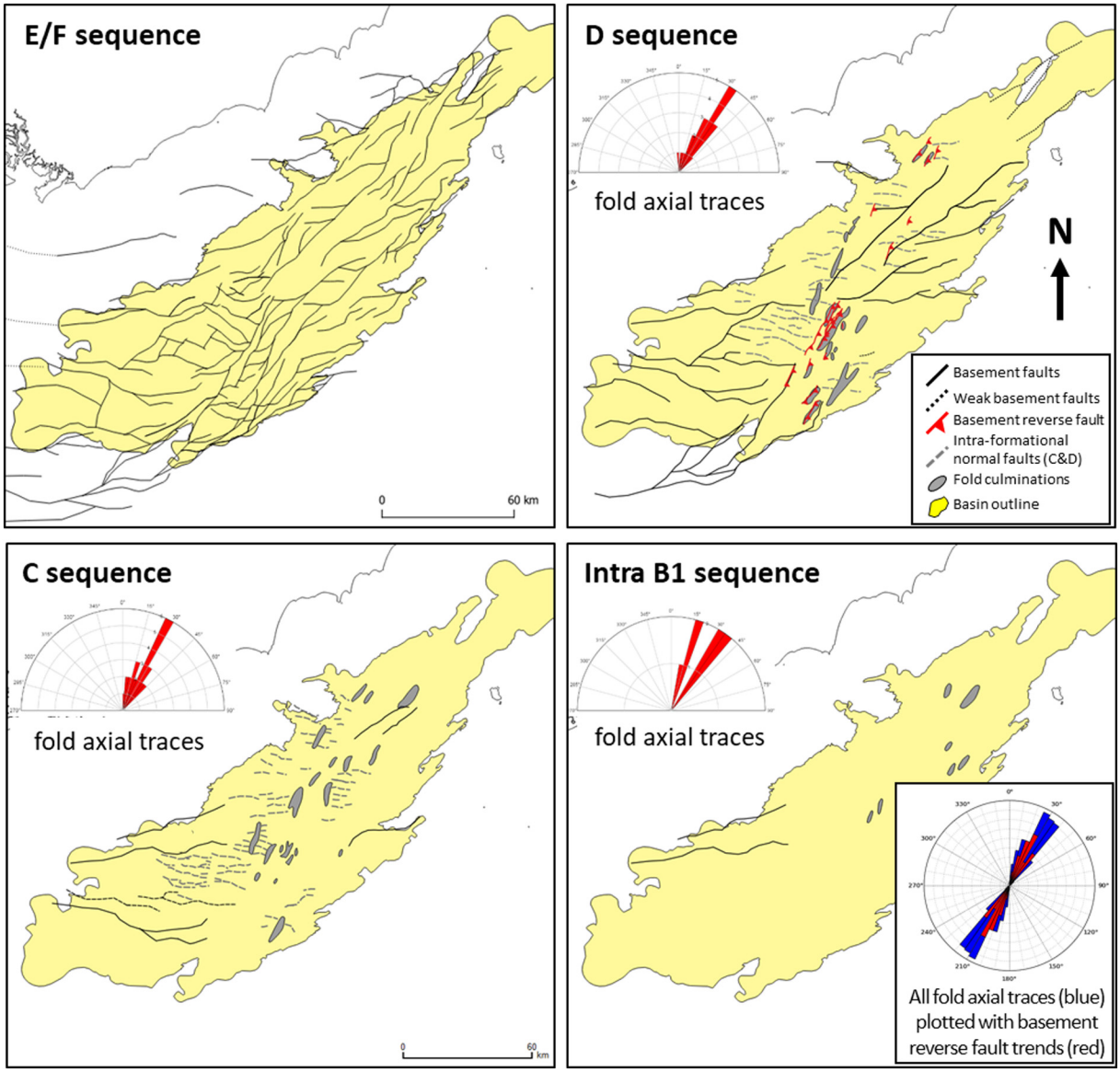


Fig. 9. Maps showing structures that were active during the E/F (combined), D, C and Intra B1 sequences. The basin outline shows the primary depocenter for each sequence. Fault traces show basement faults that extend into each of the overlying sequences. The fold culminations are shown in the D, C and B1 sequences as well as faults. Note that the intra-formational normal faults in the D and C sequence form at right angles to the adjacent folds. The overall basin outlines for all the sequences strike NE.

Fig. 13C illustrate the decrease in folding in the uppermost D sequence.

Typically, the E and F sequences are not folded, because none of the deep half-grabens were inverted enough to cause them to be folded. The lack of folding in the E/F may also partly relate to intense lithification of those sequences prior to compression.

WNW striking strata-bound normal faults in the upper D and C sequences formed during compression and inversion. These normal faults are confined to the D, C, and lowermost B1 sequences, they tip-out upward and downward, and they are oriented perpendicular to the compressional fold axes (Figs. 9 and 13). They formed at the same time as the folding in the D and C sequences (Fig. 13C and D), indicating that WNW-trending compression was accompanied by a small amount of NNE-trending extension during uppermost D and C deposition. The upper WNW normal fault tips may be partly due to depositional overlap of the faults, but the lower fault tips appear to be mechanical in origin because they terminate in the shale-rich lower portion of the D sequence (Fig. 13C and D).

The fold orientation in the D and C sequences is strongly influenced by the trend of underlying basement faults. Nearly all the fold axial traces in these sequences follow the trend of underlying basement faults with similar trends.

4.3.3. Thermal sag

We correlate the beginning of the thermal sag phase to the end of compressional folding in the lowermost B1 sequence (~25 Ma) and the onset of broad, systematic, regional subsidence (Figs. 3 and 6). The thermal sag sequences consist of mostly unfaulted basin fill sediments that thicken into the center of the basin (Fig. 6). Deposition progressed from continental at the base through transitional marine to open marine at the top in an overall coarsening upwards succession (Fig. 3). The change in depositional environment from dominantly continental to dominantly marine occurred during a major marine transgression at the top of the B1 sequence (VPI unpublished studies).

Some reactivation of older normal faults is observed in the thermal sag sequences. These faults most often form near the crests of the large basement highs in the central portion of the basin. Examples can be

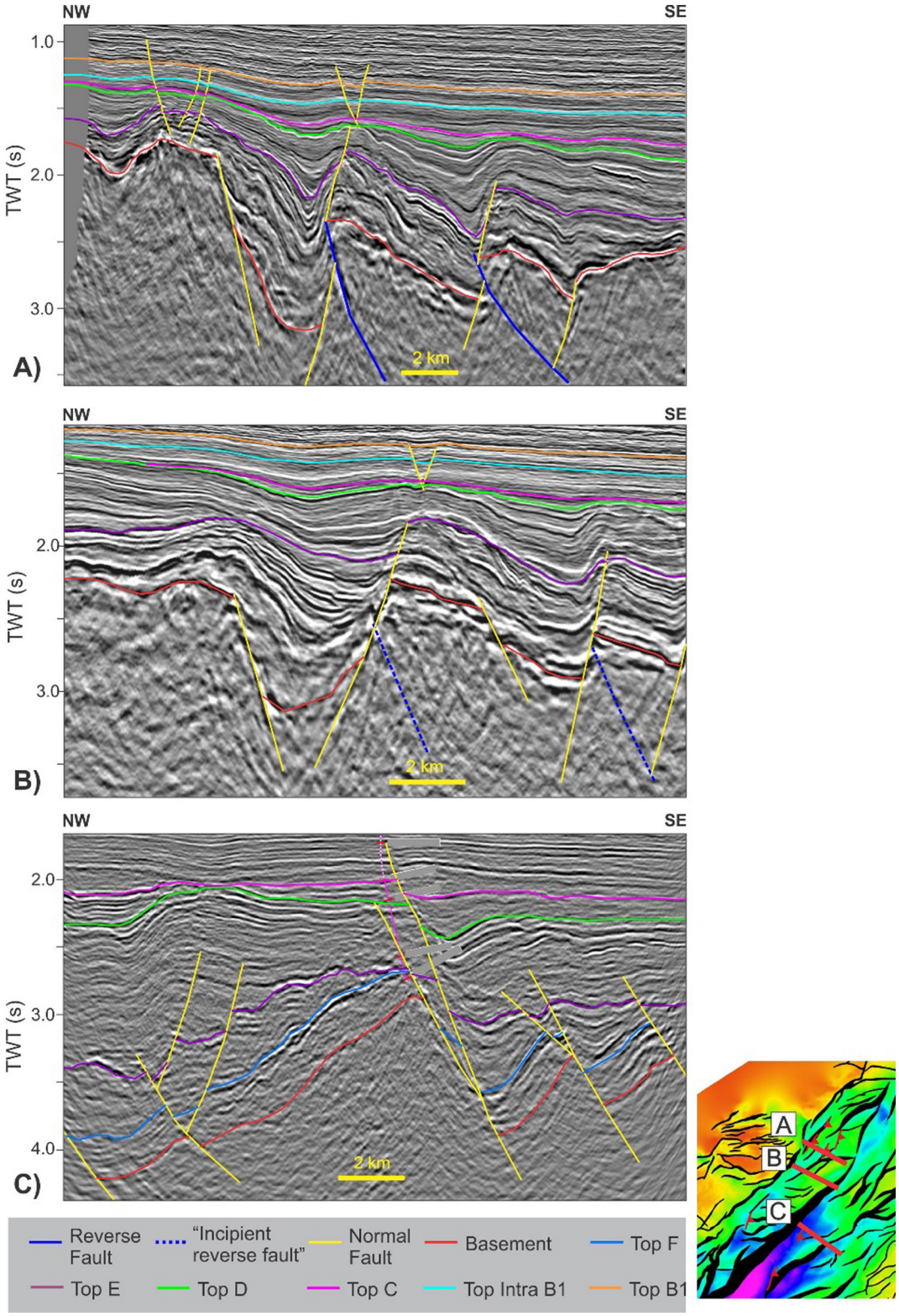


Fig. 10. 3D seismic sections over Su Tu Nau (A and B) and Su Tu Trang fields. Section A shows offset of older normal faults shown in Section B. All of the sections illustrate onlap of the D sequence onto the top E sequence unconformity. Section C shows footwall erosion of the E sequence. The E sequence conformably overlies the F sequence and a thick D sequence laps onto the top of the E sequence forming an angular unconformity in the area of the footwall uplift. Section B shows growth of the upper D sequence across NW dipping normal faults, indicating rifting extended into the upper D sequence. The inversion feature shown on Section C can be dated as uppermost D based onlap of the C sequence on the uplifted fold. Thus, inversion occurred shortly after the end of rifting. Location of inset map is shown in Fig. 4.

seen at Su Tu Trang (Fig. 10C) and at Bach Ho fields, where the intensity of differential compaction faulting during thermal sag subsidence increases from the northern flank (Fig. 11A) to the crest (Fig. 11C) of the field. Differential compaction occurs adjacent to large basement highs where thick sequences of initially high porosity sediments are juxtaposed against relatively incompressible basement.

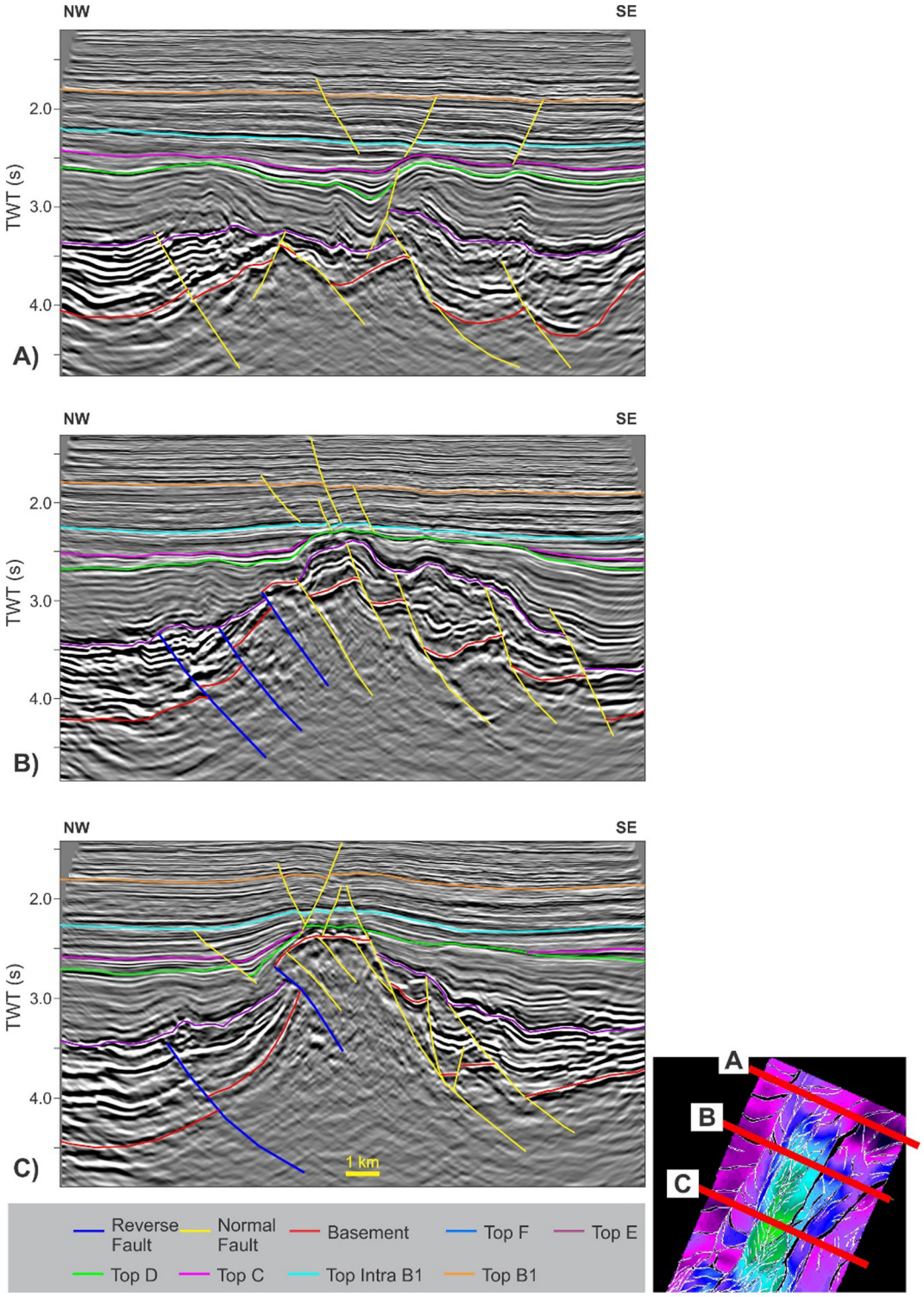


Fig. 11. Seismic sections from a high-quality, 4-component 3D seismic survey (acquired in 2015 with multi-component seafloor receivers) over Bach Ho Field. Section A is just NE of Bach Ho Field and shows less offset on the top of E than the sections to the SW. The age of compressional folds and inversion on these sections is uppermost D sequence on based on the growth of folds during deposition of the upper D sequence (on all 3 sections) and the thrusting of basement over the upper D sequence seen on the Section C. Location of inset map is shown in Fig. 4.

Upward propagation of normal faults due to differential compaction does not require additional extension.

4.4. Estimated crustal extension

A restored cross section across the Cuu Long Basin through Bach Ho Field illustrates the tectonic development of the basin (Fig. 14). The main phase of deformation is extensional followed by a smaller amount of contractional deformation and inversion on earlier normal faults. 3D seismic data coverage now allows for a more accurate description of both phases of deformation. Fig. 14 corresponds to seismic line A in Fig. 6 and line 6 in Fig. 15. It was chosen for restoration because it

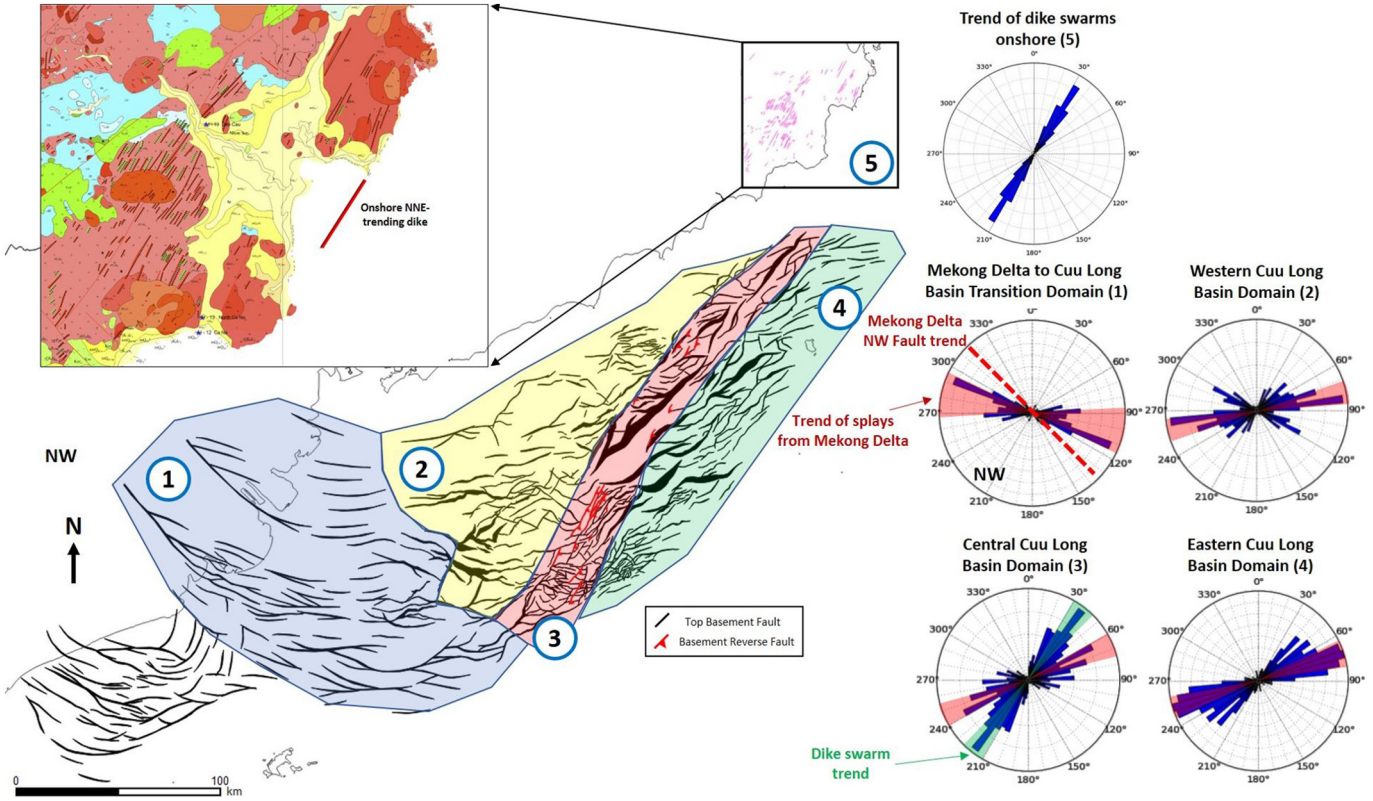


Fig. 12. Top basement fault trends in 4 structural domains in the Cuu Long Basin. The trends of splays from the Mekong Delta (red high-light) curve counter-clockwise as they propagate to the NE. Paleocene to mid Eocene dikes mapped onshore (5), along strike with the Cuu Long Basin, have a similar trend as the younger Eocene and Oligocene faults in the Central Domain (3). The geological map of dikes onshore (5) is from the General Department of Geology and Minerals of Vietnam (1998).

crosses the part of the basin that experienced the largest amounts of both extension and shortening. It shows approximately 19 km of extension followed by approximately 5 km of shortening. Using this restoration as calibration, the net crustal extension along the length of the basin was measured by summing the fault heaves on the top basement surface for 11 seismic profiles (Fig. 15). The extension measured on the top of basement is comparable to, but less than, estimates by Huchon et al. (1998) using gravity modelling. Extension measured using fault heaves on the seismic profiles and the Huchon et al. (1998) gravity modelling technique both yield the minimum value for crustal extension because they measure extension net of younger crustal shortening by inversion on normal faults. As shown in Fig. 14, the actual crustal extension in the central part of the basin is up to five kilometers more than shown in Fig. 15. It is important to note that the greatest amount of shortening occurs in the southern half of Domain 3 (Fig. 12) in the central part of the Cuu Long Basin, so only the profiles between 4 and 8 in Fig. 15 require significant corrections for shortening. These corrections yield crustal extension values of 17–21 km in the heart of the Cuu Long Basin.

4.5. Major unconformities

There are three major basin-wide unconformities in the Cuu Long Basin at the top of basement, the top of the E sequence and the top of the C sequence (Fig. 3). Throughout the basin, Mesozoic crystalline basement is nonconformably overlain by Cenozoic siliciclastics. In the heart of the basin and in the deeper grabens the F sequence is deposited on top of basement. In most other areas basement is overlain by the E sequence. Both the E and F sequences are synrift deposits with local unconformities over basement highs related to faulting and associated footwall uplift during rifting. Thus, unconformities within the E and F sequences are primarily seen on basement highs (Figs. 10C and 11C) and along the basin flanks where the E sequence onlaps the top of basement (Fig. 6).

The unconformity at the top of the E sequence is marked by onlap of the D sequence as well as large depositional hiatuses on the flanks of the basin and on structural highs (Figs. 6, 10C and 11C). A change in sedimentation also occurs at this boundary, with more sandy deposits of the E overlain by the shale dominated D sequence. The sandstones in the E sequence are much harder and denser than the overlying D shales, so this boundary has a large seismic impedance contrast and is easily correlated throughout the Cuu Long Basin.

The youngest regional unconformity occurs at the top of the C sequence. It marks the end of compressional folding and faulting in most of the basin (Fig. 9). There are a few compressional features mapped in the northeast portion of the basin that deform the lower B1 sequence, but those folds are weakly developed (Fig. 9). Pronounced thermal subsidence followed deposition of the C sequence. This subsidence created accommodation space for the B1 sands and shales, which onlap the top of the C sequence (Fig. 6).

5. Discussion and regional tectonic implications

This work refines the understanding of the formation of the Cuu Long Basin, but many of the observations are consistent with previous authors (e.g., Lee et al., 2001; Schmidt et al., 2003; Hung and Hung, 2004; Cuong and Warren, 2009; Fyhn et al., 2009; Matthews et al., 1997). The extensive data set used for this evaluation (Fig. 2) allows better understanding of the complexities of Cuu Long Basin rifting, more accurate identification of where shortening and inversion occurred, and better interpretation of the relationship between the deformation in the Cuu Long and Vinh Chau basins with strike-slip

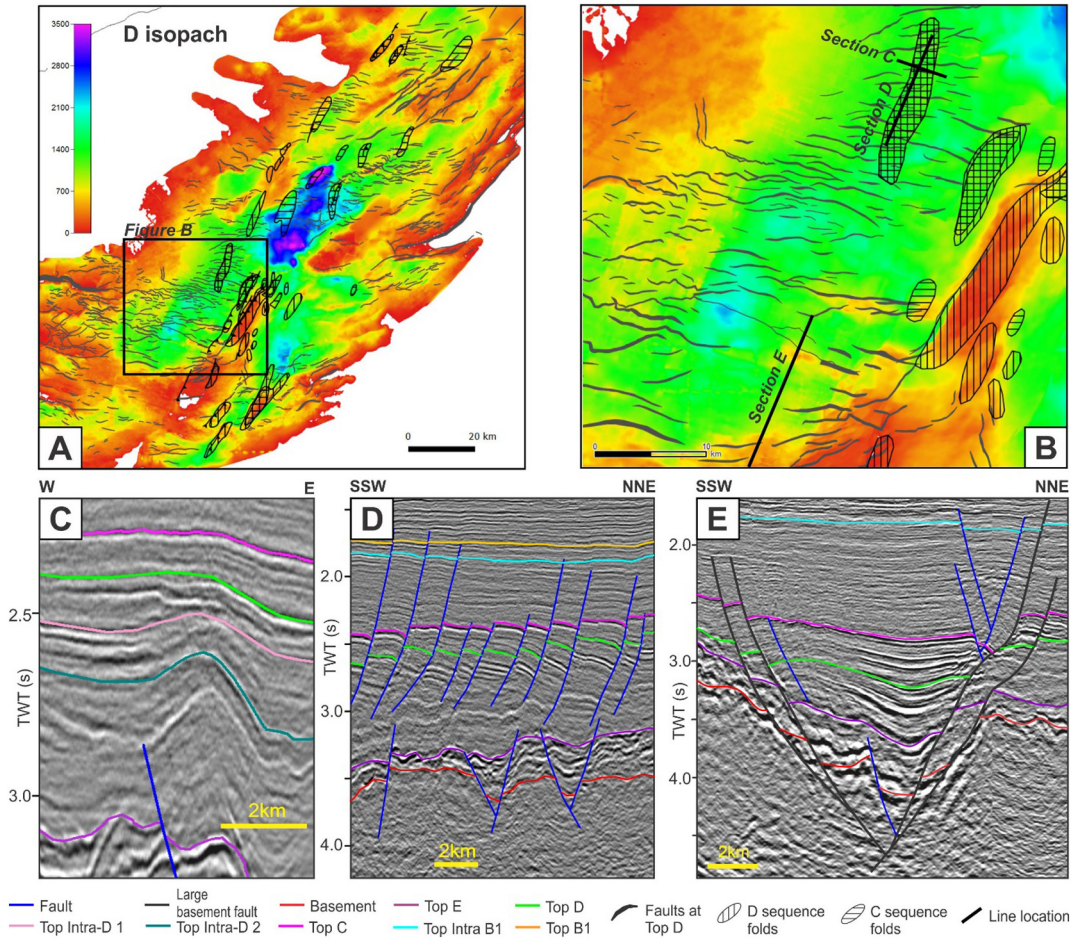


Fig. 13. Map A shows the D sequence isopach with overlays of faults and folds that formed during deposition of the D and C sequences. Inset Map B (ref. Map A) shows the location of the 3 seismic sections. The 3D seismic sections illustrate folding and extensional faulting associated with the D and C sequences. Section C shows a compressional fold that formed in the upper D sequence and early C. The intra D and C sequence faults on Section D are the same age as the fold. Section E shows D sequence and C sequence extensional faulting that extends to the basement.

faulting.

5.1. Deformational events

5.1.1. Pre-existing zones of weakness

Prior to rifting in the Cuu Long Basin, at least two NE-trending fabrics formed. The basement of southeastern Vietnam and the adjacent offshore is part of a NE-trending Late Jurassic to Cretaceous, dominantly granitic, igneous arc. Fyhn et al. (2010), Shellnutt et al. (2013), and Hamilton (1979) linked this arc to the presence of a NE-striking subduction zone farther offshore to the southeast. The NE-trending granitic arc rocks are intruded by NNE-striking Paleocene to Early Eocene mafic dike swarms (Fig. 12; Bao et al., 2001). It is likely that the NNE-striking mafic dike swarms were intruded into dilational zones of weakness in the NE-striking granitic belt associated with left-lateral, arc-parallel strike-slip during the late stages of arc magmatism analogous to the right-lateral strike-slip deformation in the northern Sumatra arc today (Muksin et al., 2014) or by post-orogenic extension in the Late Cretaceous and earliest Tertiary (Shellnutt et al., 2013).

The same NE and NNE trends are seen in normal fault trends in the central domain of the Cuu Long Basin (Domain 3; Fig. 12). The NE trend of the Jurassic to Cretaceous igneous arc is parallel to the deep axis of the Cuu Long Basin. The NNE trend of the onshore Paleocene to Early Eocene dikes parallels a significant fraction of the normal faults and almost all the compressional structures. These observations are consistent with the interpretation by Lee et al. (2001) that pre-existing zones of weakness in the Pre-Tertiary basement controlled initial rifting in the Cuu Long Basin.

5.1.2. Transtensional rifting (Eocene to mid-Early Oligocene)

Based on our seismic interpretation, nearly all the faults that cut the top of basement initially formed during rifting (Fig. 9). The interpretation also shows that all of the NE-trending normal faults in the Cuu Long and Vinh Chau basins were active at the same time and appear to be linked to near-shore and onshore NW-trending faults (Fig. 4). None of the NW-trending faults cut across the Con Son Swell. This means they all terminate into the extensional Cuu Long Basin, as originally proposed by Tapponnier et al. (1986). It also means the amount of crustal extension in the basin (Fig. 15) is equivalent to the amount of left-lateral displacement that occurred along the NW trending strike-slip fault zone onshore (Fig. 16).

Faults that were active during the rifting phase in Cuu Long Basin are shown in Fig. 16. The ESE-striking splays off the NW-trending faults in the Mekong Delta Fault Zone fit the Kim and Sanderson (2006) definition of extensional synthetic splay faults. The trend of these faults curves counter-clockwise from the NW-trending strike-slip faults in the Mekong Delta Fault Zone to synthetic splays with a trend of ENE before they link into the more common NE and NNE normal faults (Figs. 12 and 16). The deepest part of the basin is in the central domain (Domain 3, Fig. 12), where large-offset NE to NNE normal faults form horsts and grabens. Horsts in this part of the basin form the main producing oil and gas fields while the adjacent grabens are the primary hydrocarbon

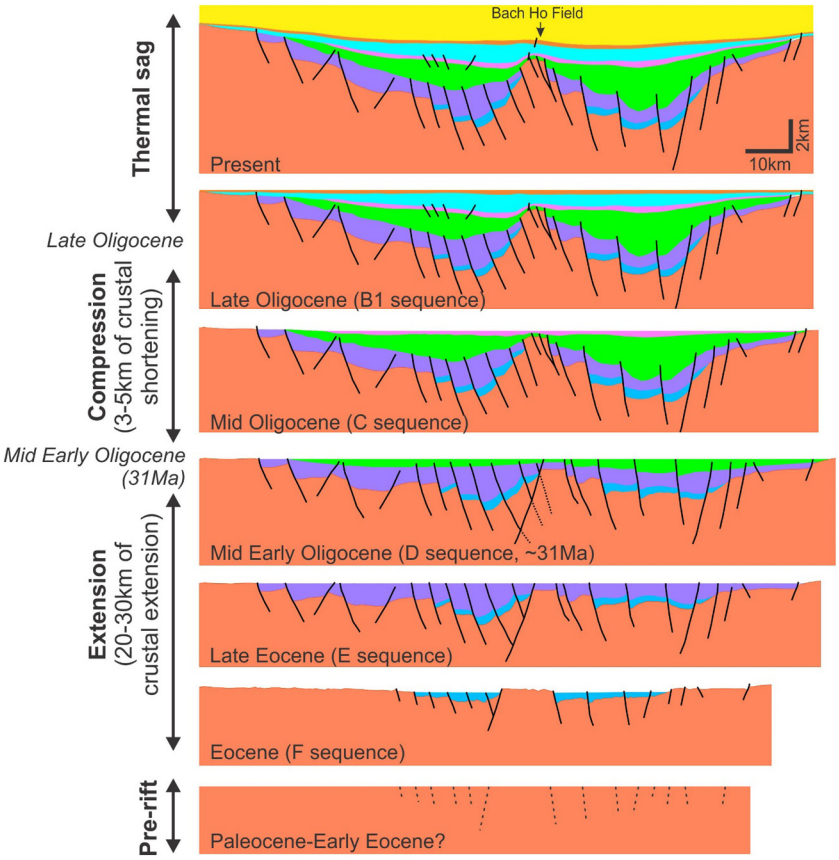


Fig. 14. Restoration of seismic Section A (across Bach Ho Field) from Fig. 6 that illustrates the tectonic evolution of the Cuu Long Basin from Eocene to present. The 3 tectonic regimes are indicated. Inferred zones of weakness in the pre-rift basement are shown. The maximum amount of extension (19 km) occurred at 31 Ma, just prior to the onset of compression. The present-day extension is 14 km, indicating 5 km of crustal shortening after rifting.

generation sites. The NNE-striking faults have the same trend as Paleocene to mid-Eocene dikes onshore (Figs. 12 and 16).

The orientation of the curved faults in the Cuu Long Basin relative to the very straight Mekong River channels, which are inferred to be along splays of the Mekong Delta Fault Zone, is similar to extensional horsetail splays described by Cunningham and Mann (2007) and Kim and Sanderson (2006). All of the linear NW trending faults near shore and the onshore Mekong River tributaries either enter or trend toward the southern third of the Cuu Long Basin. The maximum extension in the Cuu Long Basin occurs adjacent to the NE edge of where the NW trending Mekong Delta fault segments enter the basin (Figs. 6 and 15). This suggests that strike-slip displacement in the Mekong Delta Fault Zone occurred along several parallel strike-slip faults, with the maximum extension in the basin corresponding to the point in the basin with the largest sum of left-lateral strike-slip offset.

In the zone of fault curvature between the NW-trending Mekong Delta Fault Zone and the main Cuu Long Basin normal faults, the amount of dip-slip increases as the faults curve counter-clockwise to the NE. In the central part of the basin the NE-striking faults almost exclusively have dip-slip displacement. Mouslopoulou et al. (2007) describe the transfer between intersecting orthogonal strike-slip and extensional fault systems in a similar way, where the strike-slip system curves into the extensional system along intermediate oblique-slip transtensional faults.

The Siberian Baikal Rift appears to be a good analogue for the Cuu Long Basin (Kim and Sanderson, 2006). Both basins formed at about 90°

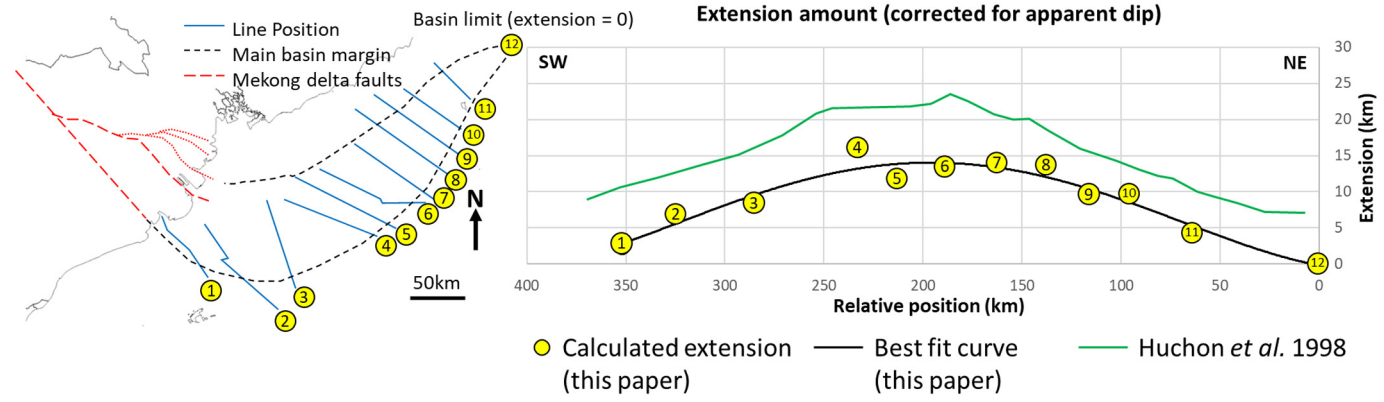


Fig. 15. Crustal extension estimates have been calculated based on fault heaves at the top basement along 11 seismic profiles in the Cuu Long Basin. For reference, the restored cross section in Fig. 14 corresponds to Line 6 above. Fault heaves on profiles not oriented in the optimum NW-SE fault dip orientation were too large, so they were corrected using the apparent dip equation and the average fault dip azimuth. The Huchon et al. (1998) crustal extension estimate, based on 3D gravimetric modelling, is shown for reference. The calculated extensions shown in this figure are not corrected for crustal shortening following rifting. Fig. 14 estimates the total crustal extension for line 6 to be 19 km, followed by 5 km of crustal shortening.

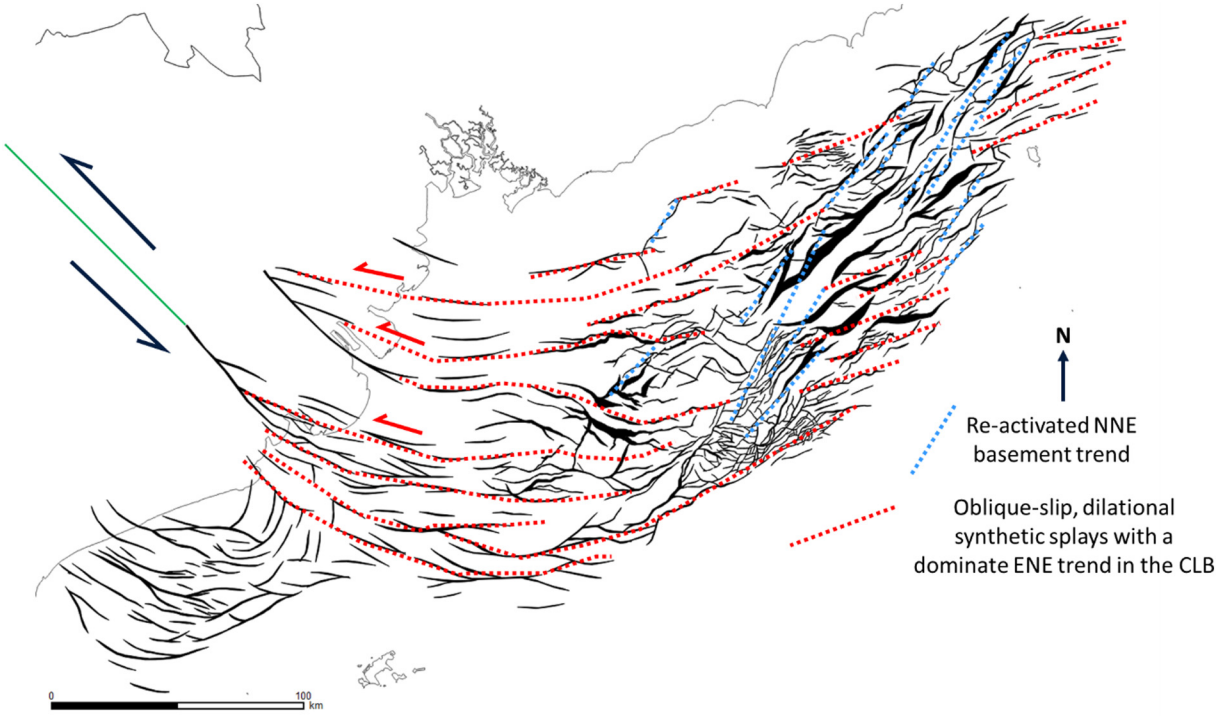


Fig. 16. Faults at top of basement and sense of shear during rifting. In the Cuu Long Basin NE and ENE trends dominate. The NNE trends are most likely related to pre-existing structures in the basement that were reactivated during rifting. The trend of dilatational synthetic splays faults curve counter-clockwise as they propagated to the NE into the Cuu Long Basin. The synthetic splays trending WNW had dominantly strike-slip displacement during rifting. The fault trends in the Vinh Chau Basin show the same sense of counter-clockwise rotation.

to the trend of the controlling strike-slip fault. The orthogonal orientation between extension and strike-slip displacement suggests that the strike-slip displacement is parallel to the direction of extension. Kim and Sanderson (2006) characterized both the Cuu Long Basin and Baikal Rift in Siberia as examples of strike-slip faults that terminate in extensional basins. Both basins have faults that curve and transition from strike-slip shear into the extensional rift basin. The transition zone faults in the Cuu Long Basin are more smoothly curved than in the Baikal Rift, but in both basins they can be described as transtensional synthetic splay faults. Both the Baikal Rift and Cuu Long Basin also reactivate pre-existing structural trends. The NE-striking Baikal Rift lies on the boundary between the Siberian Craton and Amurian Plate (Logatchev and Zorin, 1992; Petit et al., 1996). Rifting in the Cuu Long Basin reactivated a NE Late Jurassic to Cretaceous arc trend and the NNE trend of the onshore mafic dike swarms (Fig. 12).

Multiple fault trends existed in the Cuu Long Basin following the rifting event (Fig. 16). Reches (1978, 1983) and Aydin and Reches (1982) note that only four orthorhombic fault sets are required to accommodate general, three-dimensional strain (Krantz, 1988; Healy et al., 2015). Thus, the multidirectional faults cutting the basement at the end of rifting (Fig. 16) were sufficient to accommodate subsequent strain without forming new fault trends.

5.1.3. Transpression (mid-Early Oligocene to mid-Late Oligocene)

Of the multiple fault trends that formed during the rifting event, the NNE trend was preferentially reactivated during transpression (Fig. 9). The change from rifting to transpression occurred quite abruptly during latest D sequence deposition around 31 Ma and is best explained by a reversal in the sense of strike-slip from left-lateral to right-lateral displacement on the Mekong Delta Fault Zone. During rifting, the NW-striking Mekong Delta Fault Zone was left-lateral (Fig. 16). A switch to right-lateral motion would have changed the strain in the basin from extensional to contractional, creating compressional folds and reverse faults. NNE-trending normal faults during extension become NNE-trending reverse faults during right-lateral displacement in the Mekong Delta Fault Zone.

Fig. 13 shows the D isopach with superimposed reverse faults, compressional folds and WNW-striking syn-compression normal faults. The compressional folds occur in a right-stepping, en echelon pattern. The synchronous intraformational normal faults formed perpendicular to the axial traces of the folds (Fig. 13B–D). These NNE-trending folds and the associated WNW-striking normal faults formed predominantly within the upper D and C sequences at the same time as reverse slip occurred on older NNE-striking normal faults (Figs. 9 and 13). Almost all the contractional structures follow the NNE trend of onshore Paleocene to mid-Eocene dike swarms (Figs. 9 and 12).

5.2. Regional tectonic setting

The regional context of the Cuu Long Basin relative to key tectonic events in Indochina and the adjacent South China Sea have been summarized in Fig. 17. Quaternary sediments cover nearly all of southernmost Vietnam and adjacent Cambodia (Fig. 4) making it impossible to directly link NW-striking faults in the Mekong Delta Fault Zone to the Mae Ping Fault Zone further northwest. The approach taken here was to use regional geology, satellite images, topographic features, historical earthquake data and gravity data to constrain the linkage between the Cuu Long Basin, Mekong Delta Fault Zone, and the Mae Ping system. An important aspect of this analysis was to assess the impact that pre-existing structures played in linking the Mekong Delta and Mae Ping faults zones.

5.2.1. Mae Ping Fault Zone

The Mae Ping and Mekong Delta fault zones share similar orientations, displacement timing, and sense of shear histories during the Eocene to Early Miocene (Fig. 17). Both fault zones had Eocene to Early Oligocene sinistral displacement followed by dextral strike-slip in the mid Oligocene to Early Miocene. The timing of left-lateral slip is

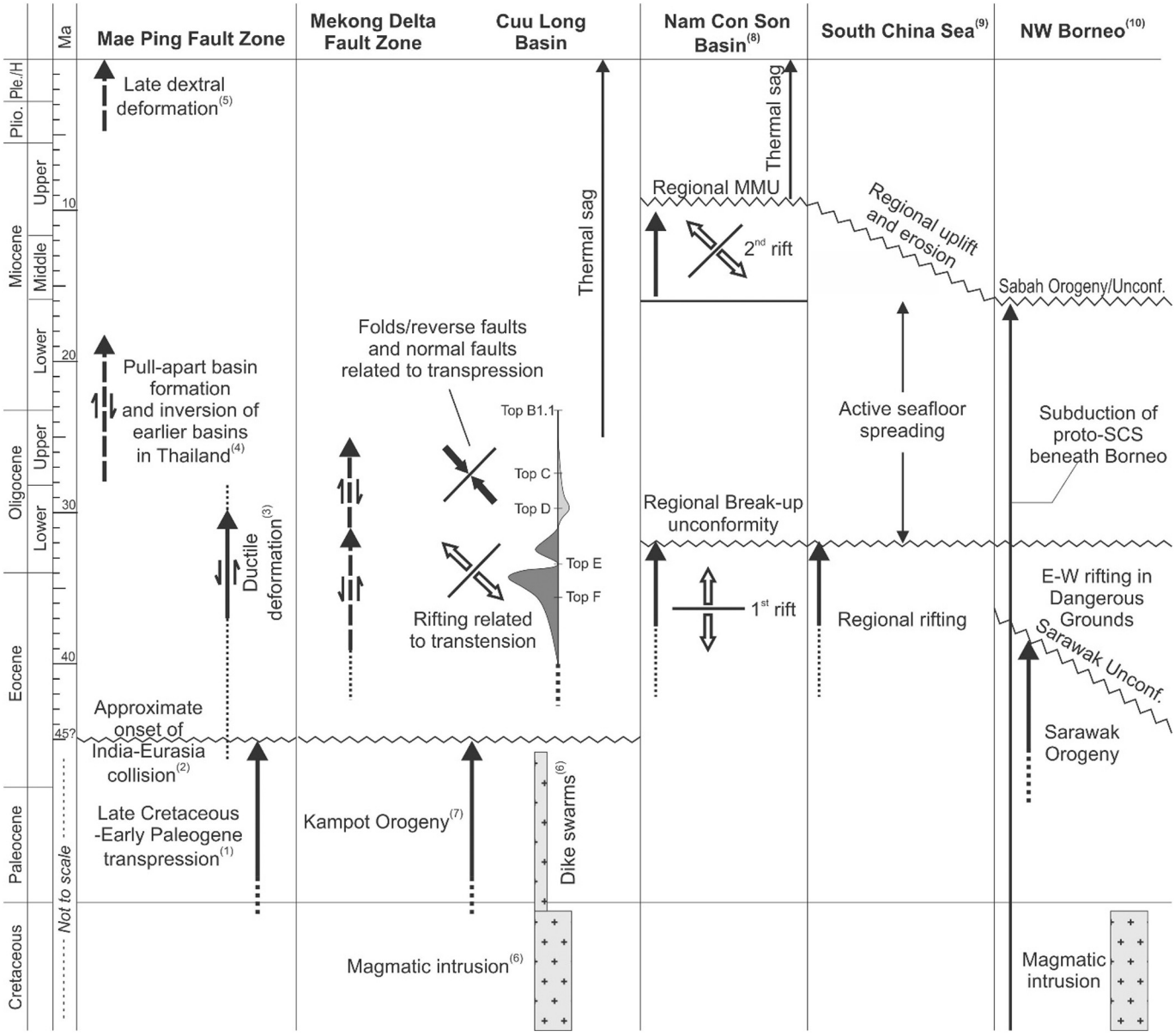


Fig. 17. Tectonic events chart for the Cuu Long Basin and adjacent areas. The Mekong Delta Fault Zone has a similar displacement history as the Mae Ping Fault Zone during the Eocene to Early Miocene. The Nam Con Son Basin and Cuu Long Basin, which are separated by the rigid Con Son Swell, have markedly different tectonic histories. Ages for the Top F to Top B1.1 in the Cuu Long Basin from R. Morley et al. (under review). References: (1) Morley (2004, 2012). (2) Tapponnier et al. (1986), Huchon et al. (1994), Lacassin et al. (1997), Leloup et al. (2001). (3) Lacassin et al. (1997), Palin et al. (2013), Smith et al. (2007), Morley et al. (2011). (4) Smith et al. (2007), Morley et al. (2011). (5) Morley et al. (2011). (6) Fyhn et al. (2010), Bao et al. (2001). (7) Fyhn et al. (2010, 2016). (8) Matthews et al. (1997). (9) Briais et al. (1993); Savva et al. (2014), Morley (2016). (10) Hutchison (1996, 2010), Moss (1998), Madon et al. (2013).

documented by radiometric age dating of ductile deformation fabrics and offsets along the Mae Ping fault zone. Lacassin et al. (1997) used Ar/Ar radiometric age dates of metamorphic rocks to conclude that ductile left-lateral deformation along the Mae Ping Fault lasted from 40 Ma until 30.5 Ma, with brittle left-lateral shearing probably continuing for a short time afterward before switching to right-lateral displacement. Morley (2004, 2012) notes that left-lateral displacement may have started much earlier during a Late Cretaceous-Paleogene transpressional episode along the western margin of Indochina, before the India-Eurasia collision started. Palin et al. (2013) used Th-Pb age dating of monazite from a sheared orthogneiss and an undeformed granite dyke along the Mae Ping Fault Zone to support Eocene (45–37 Ma) deformation and metamorphism that could have involved slip along the fault zone. Palin et al. (2013) also suggested this timing supported Late Cretaceous-Paleogene transpression and provides a maximum age for the inception of exhumation of deep crustal metamorphic rocks.

The structural history of the Mae Ping Fault Zone after 30 Ma is less well constrained. Lacassin et al. (1997) made the link between the apparently continuous penetration of India into Eurasia and activity along the Mae Ping Fault Zone, and thus assumed continuous right-lateral displacement from 30 Ma to present. Subsequent geological studies have shown a much more complicated deformation history. The formation of small pull-apart basins situated at right-stepping releasing bends, as well as other adjacent rift basins in northern Thailand, suggest that dextral motion took place shortly after sinistral motion stopped until about 18–20 Ma (Morley et al., 2007, 2011). Episodic sinistral and dextral motion along the Mae Ping Fault Zone since 18 Ma has been inferred based on episodic inversion of small rift basins adjacent to the fault zone, as well as structural analysis and AFT/ZFT cooling history in

Table 1
Comparison of Cuu Long Basin deformation with the inferred displacement on the Mekong Delta and Mae Ping fault zones. Sources: (a) This paper: based on the amount of fault heave and section restoration (line length method). The estimated 20 km of extension takes into account effects of compression not accounted for in the restoration. (b) Huchon et al., 1998: based on 3D gravimetric modelling. (c) Tapponnier et al., 1986: based on offset of geological markers. (d) Lacassin et al., 1993: based on boudin restoration. (e) Morley et al., 2011: based on the size of right-lateral pull-apart basin in onshore Thailand. (f) Smith et al., 2007: based on the lack of deep basin formation in the Chainat Duplex, lack of offset of Late Oligocene to Miocene basins overlying the strike-slip fault zone.

Area	Cuu Long Basin (CLB)	Mekong Delta FZ (inferred from the CLB)	Mae Ping FZ
Left-lateral regime	~21 km[a], 20–30 km extension[b]	~20–30 km	~160 km[c], at minimum 40–50 km[d]
Right-lateral regime	Maximum ~5 km shortening[a]	~3–5 km	km to a few 10s of km[e], a few km[f]

the Chainat Duplex region (Morley et al., 2007, 2011). Modern strike-slip motion along the Mae Ping Fault is dominantly right-lateral, based on GPS and earthquake data (Morley et al., 2011).

Timing and sense of displacement on the Mae Ping Fault Zone is remarkably similar to the that of the Mekong Delta Fault Zone and associated Cuu Long Basin (Fig. 17). Extension and transtension in the Cuu Long Basin starts in the Eocene at about 40 Ma and ends at around 31 Ma, which fits with left-lateral motion along the Mae Ping Fault Zone from 45 Ma to 30 Ma, even though some of the left-lateral motion on the Mae Ping Fault Zone in Thailand started earlier. The transpression in the basin started during deposition of the upper D sequence (mid-Early Oligocene, about 31 Ma) and continued to the mid-Late Oligocene lower B1 sequence, approximately 25 Ma. The similar timing and sense of shear suggest a causal relationship between the Mae Ping and Mekong Delta fault zones and the Cuu Long Basin.

The estimated displacements in the Mae Ping Fault Zone (Tapponnier et al., 1986; Lacassin et al., 1997; Morley et al., 2007, 2011; Morley, 2013) are considerably larger than those in the Mekong Delta Fault Zone (Table 1). This large difference between the estimated amounts of strike-slip, up to 160 km of left-lateral slip and a few 10's of km of right lateral slip on the Mae Ping Fault Zone vs. approximately 20 km of left-lateral slip and 5 km of right-lateral slip on the Mekong Delta Fault Zone, suggests that most of the slip on the Mae Ping Fault was dissipated by deformation within the Indochina crust prior to reaching the distal end of the combined fault system. Based on mapped splays in eastern Thailand and the Ton Le Sap area, Morley (2013) suggested that fault splays and extensional basins caused the displacement on the Mae Ping Fault to diminish or even die out as it reaches the Ton Le Sap area. This is consistent with our estimated displacements along the Mekong Delta Fault Zone and associated Cuu Long Basin.

The mid-Late Oligocene to Recent thermal sag phase with no contraction in the Cuu Long Basin suggests that all of the youngest right-lateral motion documented on the Mae Ping Fault Zone is dissipated before reaching the Cuu Long Basin. Minor earthquake activity along the Mekong River and in the coastal area of the Mekong Delta (Fig. 18) may indicate that some slip is almost reaching the Cuu Long Basin, however there are currently no focal mechanism data available to analyze those earthquakes.

Based on the change of course of major rivers, recent earthquake data, and regional fault patterns (Morley et al., 2011; Morley, 2012, 2013), it seems likely that the Mae Ping Fault Zone extends from the Ton Le Sap area to the Mekong Delta Fault Zone through a step-over zone that follows the NNW trend of the major rivers in the area and roughly corresponds with the eastern boundary of the Kampot Fold Belt (Figs. 18 and 19). Two recent earthquakes of M4.0 and M5.0 have occurred where the Mekong River changes course from NW in the south to NNW to the north (Fig. 18). The NNW linear course of the Mekong River continues along the Ton Le Sap River to Ton Le Sap Lake (Fig. 19).

This Mekong-Ton Le Sap step-over zone appears to be analogous to the Chainat Duplex farther to the northwest along the Mae Ping Fault Zone (Fig. 18; Smith et al., 2007; Morley et al., 2007). Both NNW-striking step-over zones link regional NW-trending strike-slip fault zones and both have very similar sizes and geometries. Both step-over zones also follow pre-existing structural fabrics. The restraining bend of the Chainat Duplex follows the trend of the a Late Cretaceous-Paleogene transpression belt in Thailand (Morley et al., 2007), while the Mekong-Ton Le Sap step-over zone follows the eastern boundary of the Paleogene Kampot Fold Belt (Figs. 18 and 19).

5.2.2. Kampot Fold Belt

During the Late Cretaceous to Early Paleocene, a major transpressional event affected a large area from Thailand to peninsular Malaysia (Morley, 2004, 2012). In Cambodia and southwest Vietnam, this event deforms pre-Tertiary rocks in the Kampot Fold Belt. Fyhn et al. (2010, 2016) define the Kampot Fold Belt as a tectonic domain characterized by up to 10+ km of Paleocene to Early Eocene uplift, folding, and reverse faulting. Onshore, crustal shortening and severe erosion in the fold belt expose Paleozoic and early Mesozoic rocks, which were thrusted over Late Jurassic to Cretaceous rocks to the west (Fig. 19). Seismic imaging of folds, reverse faults, and a regional angular unconformity of Paleogene age show that the EW compression extends a few hundred kilometers offshore in the Phu Quoc-Kampot Som Basin (VPI unpublished studies; Fyhn et al., 2010, 2016).

Fyhn et al. (2010, 2016) mapped the eastern edge of the Kampot Fold Belt along the boundary between Paleozoic to Jurassic sedimentary rocks and Late Jurassic to Cretaceous igneous rocks. However, the change in orientation of the Mekong and Cai Lon rivers (Fig. 20A) forms a line that coincides with the eastern edge of a series of well-defined en echelon structural highs within the Jurassic to Cretaceous igneous belt, suggesting that the eastern edge of the Kampot Fold Belt lies farther east and that the Paleogene deformation in the fold belt also involved the late Mesozoic igneous rocks (Fig. 20A). This newly inferred eastern boundary of the fold belt is parallel with the well-defined western boundary mapped with surface geology and seismic data by Fyhn et al. (2010, 2016). The regional residual gravity map also shows gravity lineaments that coincide with the revised eastern boundary of the Kampot Fold Belt and the western boundary of the Phu Quoc-Kampot Som Basin (Fig. 20B).

It is inferred that this revised eastern boundary of the Kampot Fold Belt forms a significant NNW trending crustal boundary that the Ton Le Sap-Mekong step-over zone follows, causing the Mekong and Cai Lon river channels to turn abruptly northward (Fig. 19). The inferred onshore strike-slip faults associated with the formation of the Vinh Chau Basin also appear to curve into and follow the boundary of the Kampot Fold Belt (Fig. 19).

5.2.3. Tectonic escape of Indochina

Escape, or extrusion, tectonics appear to be the main driver for displacement along the Mae Ping and Mekong Delta fault zones since the Eocene. Southeastward motion and strike-slip segmentation of Indochina was driven by the collision of the Indian and Eurasia plates (Molnar and Tapponnier, 1975; Tapponnier and Molnar, 1977; Tapponnier et al., 1982; Tapponnier et al., 1986) starting in the Eocene (Hall, 2012) and continuing to present (Trinh et al., 2012; Wang et al., 2017). Huchon et al. (1994) show older left-lateral displacement on the Mae Ping Fault followed by younger right-lateral motion that is linked to rotation of the regional stress field in Indochina as the Indian indenter migrated northwards. Lacassin et al. (1997) also note the

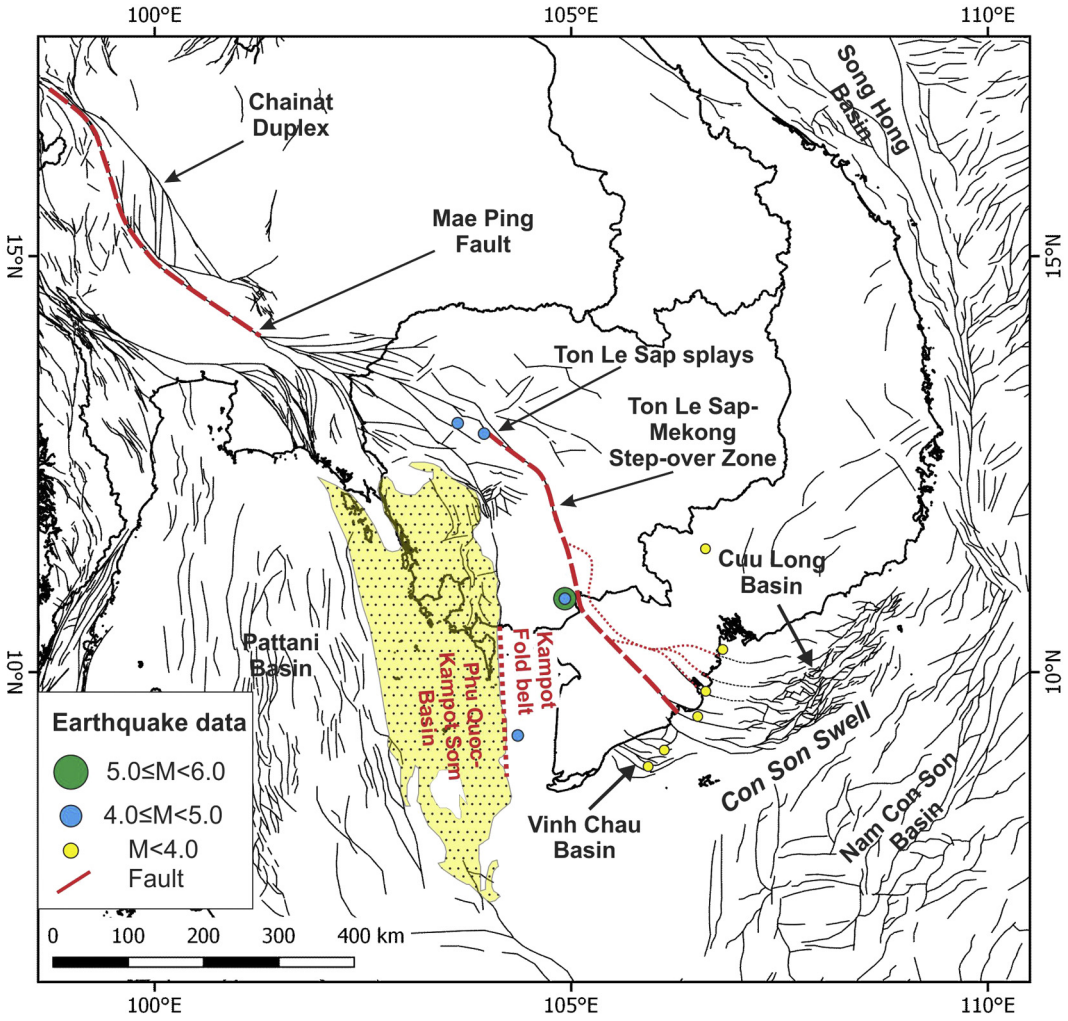


Fig. 18. The western edge of the Chainat Duplex is similar to the inferred step-over zone from the Mekong Delta Fault Zone in Vietnam to the Ton Le Sap area in Cambodia (Ton Le Sap–Mekong step-over zone). Faults onshore, in the Pattani Basin, and the outline of the Phu Quoc–Kampot Som Basin are from Morley et al. (2011) and Morley (2012, 2013). The Kampot Fold Belt boundary is from Fyhn et al. (2010). Faults for basins offshore of Vietnam are from Vietnam Petroleum Institute seismic mapping. Also shown are Earthquake foci (1900–2005) from Dung (2007).

northwards younging of left-lateral extrusion related to the northward migration of the Indian indenter, with onset of left-lateral faulting in the Mae Ping Fault Zone in Thailand beginning during the Eocene and ending at about 30 Ma followed by onset of right lateral faulting at about 23 Ma. van Hinsbergen et al. (2011) modelled left-lateral motion on the Mae Ping Fault with more constraints than Huchon et al. (1994), but their model does not show the younger right-lateral displacement. Our analysis shows the timing of the reversal of motion in the Cuu Long Basin occurred during the mid-Early Oligocene at about 31 Ma (Fig. 17).

Left-lateral strike-slip on the Mekong Delta Fault Zone is constrained by extension in the Cuu Long Basin between ~40 Ma and 31 Ma (Fig. 17) with an abrupt change to right-lateral strike-slip in the Mekong Delta Fault Zone, that caused shortening and inversion in the Cuu Long Basin, at approximately 31 Ma. The change from left-lateral to right-lateral strike-slip is also documented on the Mae Ping Fault at the same time and with a greater amount of offset (Table 1; Morley et al., 2011), which implies the driver for the change in sense of strike-slip was to the west, not the east.

The timing of the switch in sense of shear along the Mae Ping and Mekong Delta fault zones roughly corresponds to when the trailing edge of Indian continental crust passed north of the NW end of the Mae Ping Fault Zone (Fig. 21). While the block south of the Mae Ping Fault Zone was still moving southeast to get out of the way of the India-Eurasia collision, after the trailing edge of Indian continental crust passed far enough north, the southern block was juxtaposed against subducting Indian Plate oceanic crust and experienced less lateral push than farther north where continental crust was still pushing against Indochina (Fig. 21). This caused the block north of the Mae Ping and Mekong Delta fault zones to move faster and induced right-lateral strike-slip. Lacassin et al. (1997) had a similar interpretation for the change to right-lateral slip along the Mae Ping Fault Zone as India moved northward, but their date for onset of right-lateral displacement was ~23 Ma, well after the transpression started in the Cuu Long Basin at about 31 Ma.

The end of transpression in the Cuu Long Basin, at approximately 25 Ma, marks the end of strike-slip displacement on the Mekong Delta Fault Zone. It probably indicates that most of Indochina was moving as a single block by that time and that the small amount of younger motion on the Mae Ping Fault Zone documented by Morley et al. (2011) dissipated before reaching the Mekong Delta.

5.2.4. South China Sea

The Nam Con Son Basin is separated from the Cuu Long Basin by the relatively unfaulted Con Son Swell basement high. None of the faults in the Mekong Delta Fault Zone or Cuu Long Basin cut significantly into the Con Son Swell. The first rifting event in the Nam Con Son Basin resulted from NS extension at approximately 32 Ma, most likely as a

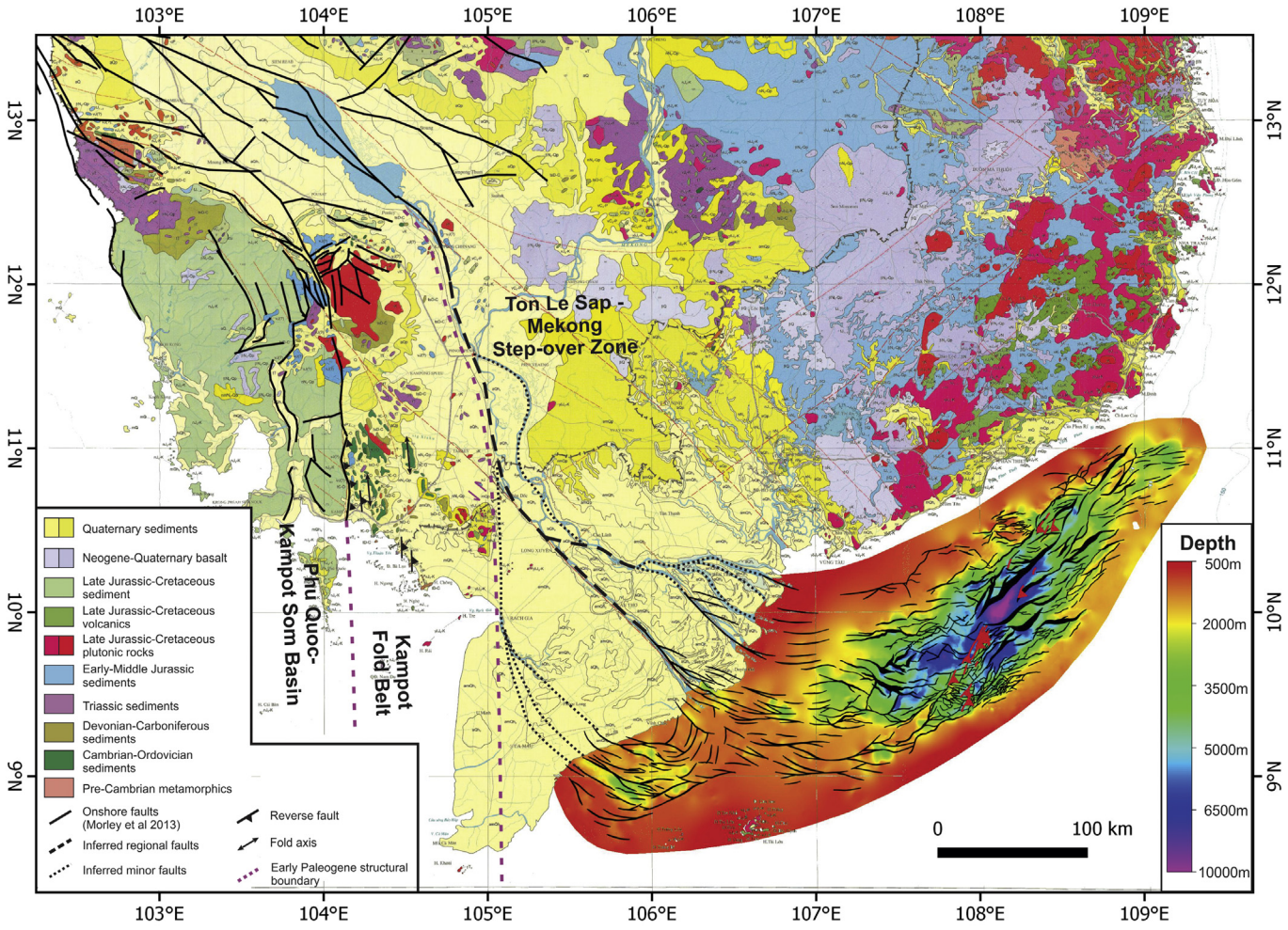


Fig. 19. Geologic map (Tien et al., 2009) showing the inferred eastern extent of the Kampot Fold Belt and inferred link of faults from the Cuu Long and Vinh Chau basins to the Ton Le Sap area via the Ton Le Sap–Mekong step-over zone. The step-over zone follows the eastern boundary of the Kampot Fold Belt from Ton Le Sap Lake to the Mekong Delta tributaries.

pre-cursor to initiation of seafloor spreading in the South China Sea (Fig. 17; Matthews et al., 1997). Almost all the extension in the Cuu Long Basin predates this earliest phase of extension in the Nam Con Son Basin and the NS extension in the Nam Con Son Basin is not seen in the Cuu Long Basin. In addition, there is no evidence that the Cuu Long Basin was affected by the main, Middle Miocene, NW-SE rifting event in the Nam Con Son Basin. Together these indicate the Con Son Swell formed a major tectonic boundary isolating the Cuu Long Basin from rifting and seafloor spreading in the South China Sea (Hutchison, 1989, 2004; Matthews et al., 1997; Huchon et al., 1998; Fyhn et al., 2009) and preventing strike-slip along the Mae Ping–Mekong Delta fault system from reaching the Nam Con Son Basin (Morley, 2002).

6. Conclusions

The dataset used for this paper is much more extensive than was available for previously published papers and allows a more detailed description of the Cuu Long Basin and its history. It confirms or only slightly modifies many of the previously published ideas about the Cuu Long Basin, especially the overall geometry and major fault patterns documented by Lee et al. (2001) and the amount of extension estimated by Huchon et al. (1998).

The major contributions of this paper are:

1. Documenting the details of the 3D architecture and deformational history of the Cuu Long Basin;
2. Identifying geometric similarities between NNE Paleocene mafic dike swarms onshore that cut the NE trending granitic arc and the NNE normal faults in the Cuu Long Basin. This dike swarm orientation is interpreted as a zone of weakness in the pre-rift basement of the Cuu Long Basin that was reactivated during the onset of rifting during the Eocene.
3. Documenting details about the Mekong Delta Fault Zone and identifying a linkage with the Mae Ping Fault Zone in Thailand and Cambodia through a right step along the eastern edge of the pre-existing Kampot Fold Belt.

The NE trending Cuu Long Basin is located at the distal, southeastern end of the Mae Ping strike-slip fault system (also referred to as the Wang Chao Fault). The basin formed in response to up to 21 km of extension between approximately 40 Ma and 31 Ma. It was subsequently shortened by up to 5 km between 31 Ma and 25 Ma. The overall basin orientation is parallel to a Late Jurassic to Cretaceous granitic arc, but many of the extensional faults in the basin strike NNE parallel to onshore mafic dike swarms that cut those same granitic arc rocks. The oldest E and F depositional sequences are synrift deposits. The late synrift and earliest post-rift D sequence onlaps the unconformity at the top of the E sequence and it contains the high TOC, high Hydrogen-Index D shale that is the primary source and seal for the large hydrocarbon accumulations in the basin (Cuong and Warren, 2009). The upper part of the D sequence fills and heals the rift basin architecture.

Shortening in the Cuu Long Basin started during deposition of the

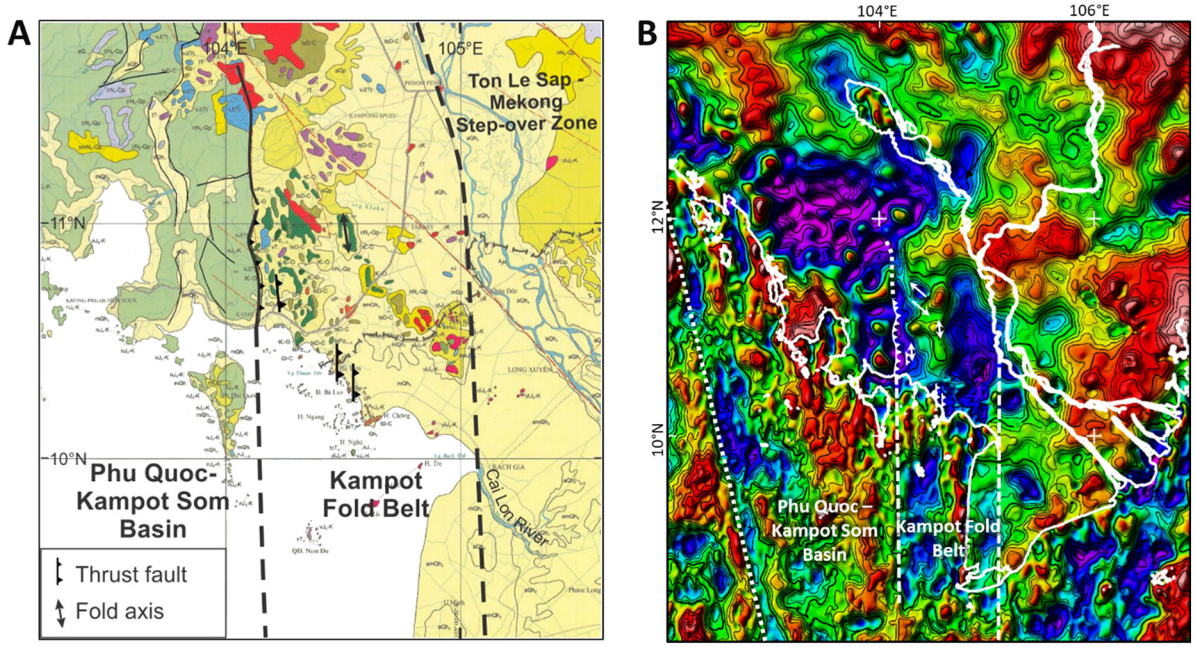


Fig. 20. The change in course of the Mekong River is similar in nature to that of the Cai Lon River to the south as shown on the geologic map (A). Both rivers change course and follow the trend of the inferred eastern boundary of the Kampot Fold Belt. A residual gravity map (processed by Mark Longacre, 2016) that merges all available marine data with onshore gravity data (B) shows NS trends on the gravity map that coincides with the eastern and western boundaries of the Kampot Fold Belt. NNW-trending Mekong and Ton Le Sap rivers also follow gravity anomalies, as does Ton Le Sap Lake. Kampot Fold Belt thrust faults and fold axes are from Fyhn et al. (2016).

youngest part of the D sequence. Contractional features are dominantly located in the central part of the Cuu Long Basin where the greatest amount of extension occurred earlier. NNE-trending contractional fold axes and reverse faults are the result of reactivated normal faults with the same trend. Some of the larger offset basement faults also have clear evidence of inversion. Intraformational WNW-striking normal faults in the D, C and lowermost B1 sequences formed to accommodate extension perpendicular to the shortening direction. The mid-Early Oligocene to mid-Late Oligocene transpression was followed by thermal sag subsidence (Figs. 3 and 17). Differential compaction-driven normal faults formed along the margins of large basement highs during the thermal sag phase, but there is no indication of additional extension.

The NW extension direction in the Cuu Long Basin (Fig. 21a) is parallel to the major tributaries in the Mekong Delta and NW trending strike-slip faults immediately offshore from the delta. Together these form the Mekong Delta Fault Zone. The Mekong Delta Fault Zone can be extrapolated northwestward along the trend of the Mekong River to the east side of the Kampot Fold Belt where the river takes an abrupt turn to the north. Morley (2013) has traced splays of the Mae Ping Fault Zone as far as the Ton Le Sap area. We postulate that the eastern margin of the Kampot Fold Belt forms a right step in the Mae Ping Fault system, called the Mekong-Ton Le Sap step-over, that connects the Mekong Delta and Mae Ping fault zones. The 21 km of maximum left-lateral slip and 5 km maximum right-lateral slip along the Mekong Delta Fault Zone are consistent with Morley's (2013) observation that fault displacement on the Mae Ping Fault Zone southeast of the Ton Le Sap area

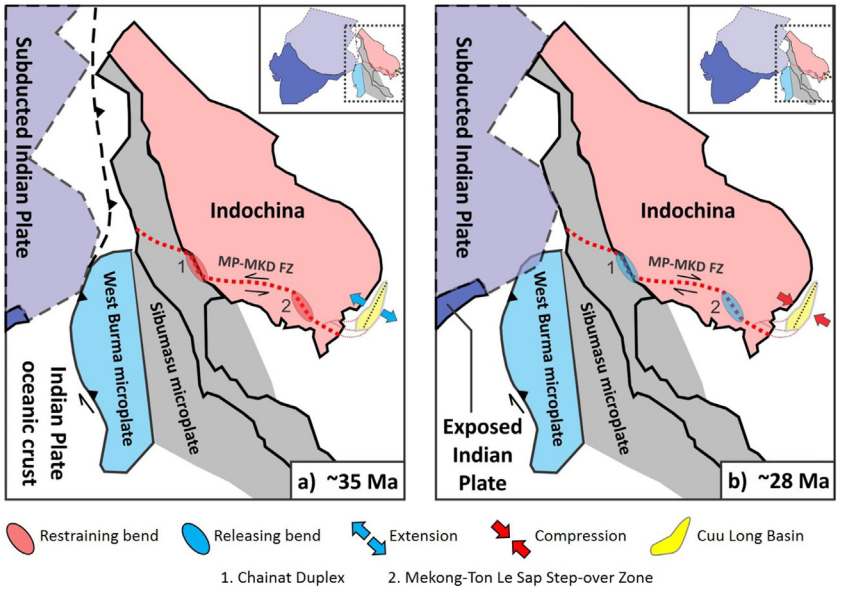


Fig. 21. Schematic reconstruction based on Gplates (http://www.gplates.org) plate boundaries and motion vectors for India, Indochina, West Burma and Sibumasu from van Hinsbergen et al. (2011). Note that the strike-slip on the Mae Ping- Mekong Delta fault system is not restored. Shape of the now subducted Indian continental crust from Hall (2012). a) NE corner of Indian Continental Crust primarily pushing on block south of the Mae Ping–Mekong Delta Fault Zone causing left-lateral strike-slip. b) NE corner of Indian Continental Crust primarily pushing on block north of the Mae Ping–Mekong Delta Fault Zone causing right-lateral strike-slip and relatively faster extrusion of the northern block.

is probably small. The Mekong-Ton Le Sap step-over would have acted as a restraining bend during left-lateral displacement (Fig. 20A), and a releasing bend during right-lateral displacement (Fig. 21b).

Finally, the NW-SE rifting in the Cuu Long Basin is unrelated to the opening of the South China Sea. Extension in the Nam Con Son Basin, at the southern end of the South China Sea, started near the end of rifting in the Cuu Long Basin at about 32 Ma and had a N-S orientation, different from the NW-SE extension orientation in the Cuu Long Basin. NW-SE rifting and seafloor spreading in the Nam Con Son Basin postdates extension in the Cuu Long Basin (Fig. 17). The Con Son Swell, which separates the two basins, is essentially unfaulted and created a major tectonic boundary separating the two basins. Although both basins were most likely moving southeastward with the extrusion of Indochina, extension in the Cuu Long Basin is directly related to Indochina being pushed out of the way of the India-Eurasia collision whereas extension in the Nam Con Son Basin is related to opening of the South China Sea ocean basin (Matthews et al., 1997).

Declaration of interest

None.

Acknowledgements

This paper would not have been possible without the support of PetroVietnam, who provided the funding and permission to publish the work as well as input on the content of the paper. Discussions with Chris Morley and Michael Fyhn were extremely helpful. Dawn Kaback provided valuable comments to multiple drafts of the paper. Reviews by Chris Morley, Michael Fyhn and an anonymous reviewer were extremely beneficial and have been incorporated into the paper. Nguyen Hong Chau prepared the high-quality satellite images used for our evaluation and included in the publication.

References

Aydin, A., Reches, Z., 1982. Number and orientation of fault sets in the field and in experiments. Geology 10, 107–112.

Bao, Nguyen Xuan, Van Binh, Nguyen, Long, Pham Huy, Van Long, Trinh, Van Linh, Do, et al., 2001. Structure and Metallogeny of Vietnam (in Vietnamese). South Vietnam Geological Mapping Division, Ho Chi Minh City.

Briais, A., Patriat, P., Tapponnier, P., 1993. Updated interpretation of magnetic anomalies and seafloor spreading stages in the South China Sea: implications for the Tertiary tectonics of Southeast Asia. J. Geophys. Res. 98 (B4), 6299–6328.

Cunningham, W.D., Mann, P., 2007. Tectonics of strike-slip restraining and releasing bends. Geol. Soc. Lond., Spec. Publ. 290 (1), 1–12. https://doi.org/10.1144/SP290.1.

Cuong, T.X., Warren, J.K., 2009. Bach Ho field, a fractured granitic basement reservoir, Cuu Long basin, offshore SE Vietnam: a 'buried-hill' play. J. Pet. Geol. 32 (2), 129–156. https://doi.org/10.1111/j.1747-5457.2009.00440.x.

Dam, M.H., Quang, C.D., 2015. Sequence stratigraphy and determination of Early Miocene–Oligocene sedimentation environments of block 09-3 in Cuu Long basin based on the characteristics of freshwater dinocyst groups and palynofacies analysis. Petrovietnam Journal (7), 24–32.

Dung, N.Q., 2007. Earthquake Activity and Seismological Network in Vietnam. Presented at the GLORIA Symposium. Annex Hall, Pacifico Yokohama, Yokohama, JAPAN.

Fyhn, M.B.W., Boldreel, L.O., Nielsen, L.H., 2009. Geological development of the Central and South Vietnamese margin: implications for the establishment of the South China Sea, Indochinese escape tectonics and Cenozoic volcanism. Tectonophysics 478 (3–4), 184–214. https://doi.org/10.1016/j.tecto.2009.08.002.

Fyhn, M.B.W., Pedersen, S.A.S., Boldreel, L.O., Nielsen, L.H., Green, P.F., Dien, P.T., ... Frei, D., 2010. Paleocene-early Eocene inversion of the Phuquoc-Kampot Som Basin: SE Asian deformation associated with the suturing of Luconia. J. Geol. Soc. 167 (2), 281–295. https://doi.org/10.1144/0016-76492009-039.

Fyhn, M.B.W., Green, P.F., Bergman, S.C., Van Itterbeeck, J., Tri, T.V., Dien, P.T., ... Nielsen, L.H., 2016. Cenozoic deformation and exhumation of the Kampot Fold Belt and implications for south Indochina tectonics. Journal of Geophysical Research: Solid Earth 121 (7), 5278–5307. https://doi.org/10.1002/2016JB012847.

General Department of Geology and Minerals of Vietnam, 1998. Geological and Mineral Resources Map of Vietnam, Scale 1:200,000.

Hall, R., 2012. Late Jurassic-Cenozoic reconstructions of the Indonesia region and Indian Ocean. Tectonophysics 570, 1–41.

Hamilton, W.B., 1979. Tectonics of the Indonesian Region. US Govt. Print. Off.

Healy, D., Blenkinsop, T.G., Timms, N.E., Meredith, P.G., Mitchel, T.M., Cooke, M.L., 2015. Polymodal faulting: time for a new angle on shear failure. J. Struct. Geol. 80, 57–71. https://doi.org/10.1016/j.jsg.2015.08.013.

Huchon, P., Pichon, X.L., Rangin, C., 1994. Indochina Peninsula and the collision of India and Eurasia. Geology 22 (1), 27–30. https://doi.org/10.1130/0091-7613(1994)022<0027:IPATCO>2.3.CO;2.

Huchon, P., Nguyen, T.N.H., Chamot-Rooke, N., 1998. Finite extension across the South Vietnam basins from 3D gravimetric modelling: relation to South China Sea kinematics. Mar. Pet. Geol. 15 (7), 619–634. https://doi.org/10.1016/S0264-8172(98)00031-2.

Hung, N.D., Hung, L.V., 2004. Petroleum geology of Cuu Long Basin-offshore Vietnam. Presented at the AAPG International Conference, Barcelona, Spain. Retrieved from. http://www.searchanddiscovery.com/documents/2004/hung/index.htmTatyana.

Hutchison, C.S., 1989. Geological Evolution of South-east Asia. Clarendon Press.

Hutchison, C.S., 1996. The 'Rajang accretionary prism' and 'Lupar Line' problem of Borneo. Geol. Soc. Lond., Spec. Publ. 106 (1), 247–261. https://doi.org/10.1144/GSL.SP.1996.106.01.16.

Hutchison, C.S., 2004. Marginal basin evolution: the southern South China Sea. Mar. Pet. Geol. 21 (9), 1129–1148. https://doi.org/10.1016/j.marpetgeo.2004.07.002.

Hutchison, C.S., 2010. Oroclines and paleomagnetism in Borneo and South-East Asia. Tectonophysics 496 (1–4), 53–67. https://doi.org/10.1016/j.tecto.2010.10.008.

Kim, Y.-S., Sanderson, D.J., 2006. Structural similarity and variety at the tips in a wide range of strike-slip faults: a review: similarity and variety at strike-slip fault tips. Terra Nova 18 (5), 330–344. https://doi.org/10.1111/j.1365-3121.2006.00697.x.

Krantz, R.W., 1988. Multiple fault sets and three-dimensional strain: theory and application. J. Struct. Geol. 10, 225–237.

Lacassin, R., Leloup, P.H., Tapponnier, P., 1993. Bounds on strain in large Tertiary shear zones of SE Asia from boudinage restoration. J. Struct. Geol. 15 (6), 677–692. https://doi.org/10.1016/0191-8141(93)90055-F.

Lacassin, R., Maluski, H., Leloup, P.H., Tapponnier, P., Hinthong, C., Siribhakdi, K., Charoenravat, A., 1997. Tertiary diachronic extrusion and deformation of western Indochina: structural and 40Ar/39Ar evidence from NW Thailand. J. Geophys. Res. 102, 10,013–10,037. https://doi.org/10.1029/96JB03831.

Lee, G.H., Lee, K., Watkins, J.S., 2001. Geologic evolution of the Cuu Long and Nam Con Son basins, offshore southern Vietnam, South China Sea. AAPG Bull. 85 (6), 1055–1082.

Leloup, P.H., Arnaud, N., Lacassin, R., Kienast, J.R., Harrison, T.M., Trong, T.T.P., ... Tapponnier, P., 2001. New constraints on the structure, thermochronology, and timing of the Ailao Shan-Red River shear zone, SE Asia. Journal of Geophysical Research: Solid Earth 106 (B4), 6683–6732. https://doi.org/10.1029/2000JB900322.

Logatchev, N.A., Zorin, Y.A., 1992. Baikal rift zone: structure and geodynamics. Tectonophysics 208 (1–3), 273–286.

Longacre, Mark, 2016, March. Southeast Asia Gravity and Magnetic Map (unpublished VPI study).

Madon, M., Ly, K.C., Wong, R., 2013. The structure and stratigraphy of deepwater Sarawak, Malaysia: implications for tectonic development. J. Asian Earth Sci. 76, 312–333.

Matthews, S.J., Fraser, A.J., Lowe, S., Todd, S.P., Peel, F.J., 1997. Structure, stratigraphy and petroleum geology of the SE Nam Con Son Basin, offshore Vietnam. Geol. Soc. Lond., Spec. Publ. 126 (1), 89–106. https://doi.org/10.1144/GSL.SP.1997.126.01.07.

Molnar, P., Tapponnier, P., 1975. Cenozoic tectonics of Asia: effects of a continental collision. Science 189, 419–426.

Morley, C.K., 2002. A tectonic model for the Tertiary evolution of strike-slip faults and rift basins in SE Asia. Tectonophysics 347 (4), 189–215. https://doi.org/10.1016/S0040-1951(02)00061-6.

Morley, C.K., 2004. Nested strike-slip duplexes, and other evidence for Late Cretaceous–Paleogene transpressional tectonics before and during India–Eurasia collision, in Thailand, Myanmar and Malaysia. J. Geol. Soc. 161 (5), 799–812.

Morley, C.K., 2012. Late Cretaceous-Early Paleogene tectonic development of SE Asia. Earth Sci. Rev. 115 (1–2), 37–75. https://doi.org/10.1016/j.earscirev.2012.08.002.

Morley, C.K., 2013. Discussion of tectonic models for Cenozoic strike-slip fault-affected continental margins of mainland SE Asia. J. Asian Earth Sci. 76, 137–151. https://doi.org/10.1016/j.jseaes.2012.10.019.

Morley, C.K., 2016. Major unconformities/termination of extension events and associated surfaces in the South China Seas: review and implication for tectonic development. J. Asian Earth Sci. 120, 62–86.

Morley, C.K., Smith, M., Carter, A., Charusiri, P., Chantraprasert, S., 2007. Evolution of deformation styles at a major restraining bend, constraints from cooling histories, Mae Ping fault zone, western Thailand. Geol. Soc. Lond., Spec. Publ. 290 (1), 325–349. https://doi.org/10.1144/SP290.12.

Morley, C.K., Charusiri, P., M Watkinson, I., 2011. Structural geology of Thailand during the Cenozoic Modern tectonic setting of Thailand. In: Geology of Thailand, pp. 273–334.

Morley, R.J., Dung, Bui Viet, Tung, Nguyen Thanh, Kullman, A.J., Bird, R., Van Kieu, Nguyen, Chung, Nguyen Hoai, 2019. High resolution Palaeogene climate and tectonic driven sequence stratigraphic framework for the Cuu Long Basin, offshore Vietnam, established from sequence biostratigraphy. Palaeogeogr. Palaeoclimatol. Palaeoecol (under review).

Moss, S.J., 1998. Embaluh Group turbidites in Kalimantan: evolution of a remnant oceanic basin in Borneo during the Late Cretaceous to Paleogene. J. Geol. Soc. 155 (3), 509–524.

Mouslopoulou, V., Nicol, A., Little, T.A., Walsh, J.J., 2007. Displacement transfer between intersecting regional strike-slip and extensional fault systems. J. Struct. Geol. 29, 100–116.

Muksin, U., Haberland, C., Nukman, M., Bauer, K., Weber, M., 2014. Detailed fault structure of the Tarutung Pull-Apart Basin in Sumatra, Indonesia, derived from local earthquake data. J. Asian Earth Sci. 96, 123–131.

Palin, R.M., Searle, M.P., Morley, C.K., Charusiri, P., Horstwood, M.S.A., Roberts, N.M.W., 2013. Timing of metamorphism of the Lansang gneiss and implications for the left-lateral motion along the Mae Ping (Wang Chao) strike-slip fault, Thailand. J. Asian Earth Sci. 76, 120–136.

Peltzer, G., Tapponnier, P., 1988. Formation and evolution of strike-slip faults, rifts, and basins during the India-Asia Collision: an experimental approach. Journal of Geophysical Research: Solid Earth 93 (B12), 15085–15117. https://doi.org/10.1029/JB093iB12p15085.

Petit, C., Déverchère, J., Houdry, F., Sankov, V.A., Melnikova, V.I., Delvaux, D., 1996. Present-day stress field changes along the Baikal rift and tectonic implications. Tectonics 15 (6), 1171–1191. https://doi.org/10.1029/96TC00624.

Reches, Z., 1978. Analysis of faulting in three-dimensional strain field. Tectonophysics 47, 109–129.

Reches, Z., 1983. Faulting of rocks in three-dimensional strain fields II. Theoretical analysis. Tectonophysics 94, 133–156. https://doi.org/10.1016/0040-1951(83)90264-0.

Replumaz, A., Tapponnier, P., 2003. Reconstruction of the deformed collision zone between India and Asia by backward motion of lithospheric blocks: blocks reconstruction of Asia. Journal of Geophysical Research: Solid Earth 108 (B6). https://doi.org/10.1029/2001JB000661.

Savva, D., Pubellier, M., Franke, D., Chamot-Rooke, N., Meresse, F., Steuer, S., Auxietre, J.L., 2014. Different expressions of rifting on the South China Sea margins. Mar. Pet. Geol. 58, 579–598. https://doi.org/10.1016/j.marpetgeo.2014.05.02.

Schmidt, W.J., P.H. Long, and N.V. Que. (2003). Tectonic development of the Cuu Long Basin, Vietnam. In Vietnam Petroleum Institute – 25 Years of Development, 87–109. Hanoi.

Shellnutt, J.G., Lan, C.-Y., Van Long, T., Usuki, T., Yang, H.-J., Mertzman, S.A., ... Hsu, W.-Y., 2013. Formation of Cretaceous Cordilleran and post-orogenic granites and their microgranular enclaves from the Dalat zone, southern Vietnam: tectonic implications for the evolution of Southeast Asia. Lithos 182–183, 229–241. https://doi.org/10.1016/j.lithos.2013.09.016.

Smith, M., Chantraprasert, S., Morley, C.K., Cartwright, I., 2007. Structural geometry and timing of deformation in the Chainat duplex, Thailand. Geol. Soc. Lond., Spec. Publ. 290 (1), 305–323. https://doi.org/10.1144/SP290.11.

Tapponnier, P., Molnar, P., 1977. Active faulting and Cenozoic tectonics of China. J. Geophys. Res. 82, 2905–2930.

Tapponnier, P., Peltzer, G., Le Dain, A.Y., Armijo, R., 1982. Propagating extrusion tectonics in Asia: new insights from simple experiments with plasticine. Geology 10, 611–616.

Tapponnier, P., Peltzer, G., Armijo, R., 1986. On the mechanics of the collision between India and Asia. Geol. Soc. Lond., Spec. Publ. 19 (1), 113–157. https://doi.org/10.1144/GSL.SP.1986.019.01.07.

Tien, P.C., Ngoc, N.L., Hung, L.Q., 2009. Geological Map of Cambodia, Laos and Viet Nam. Vietnam Institute of Geosciences and Mineral Resources.

Tran, Le Dong, Phung, Dac Hai, 2007. Cuu Long sedimentary basin and petroleum resources. The Petroleum Geology and Resources of Vietnam. In: The Science and Technology Publishing House, 1st ed. .

Trinh, Phan Trong, Van Liem, Ngo, Van Huong, Nguyen, Vinh, Hoang Quang, Van Thom, Bui, Thao, Bui Thi, Tan, Mai Thanh, Hoang, Nguyen, 2012. Late Quaternary tectonics and seismotectonics along the Red River fault zone, North Vietnam. Earth Sci. Rev. 114, 224–235.

van Hinsbergen, D. J. J., Kapp, P., Dupont-Nivet, G., Lippert, P. C., DeCelles, P. G., & Torsvik, T. H. (2011). Restoration of Cenozoic deformation in Asia and the size of Greater India: restoring Cenozoic Asian deformation. Tectonics, 30(5). doi:10/fjkj5m.

Wang, w., Qiao, X., Yang, S., Wang, D., 2017. Present-day velocity field and block kinematics of Tibetan Plateau from GPS measurements. Geophys. J. Int. 208, 1088–1102.